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Investigating Multimodal Contributions in Large Language Models Using audioDIME

TESI DI LAUREA MAGISTRALE IN
MUSIC AND ACOUSTIC ENGINEERING

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Abstract: Large Audio Language Models are getting better at answering questions about music, but correct answers alone do not prove that they are truly listening. A model may reach the right response by grounding its decision in the acoustic signal, or by relying on textual priors, music-theoretical knowledge, and biases learned during pretraining. This thesis asks whether Qwen2.5-Omni-7B actually uses sound when solving Music Question Answering tasks, or whether its decisions are mainly driven by text.

To address this question, the work builds an interpretability pipeline for multimodal music understanding. MM-SHAP is adapted to measure the relative contribution of audio and text, while audioDIME is introduced as an audio-domain variant of DIME that separates unimodal effects from multimodal interactions. The method combines logit-based scoring, Demucs source separation, onset-based temporal segmentation, and audioLIME-inspired perturbations, representing audio through interpretable source-segment pairs.

Experiments on HumMusQA reveal a clear text-first bias. Text is highly sufficient and necessary for preserving the model's decisions, whereas purely audio-based contributions are often replaceable. Audio becomes relevant mainly inside audio-text interactions, suggesting that its causal role is mediated by the question rather than acting as an independent perceptual signal.

Internal analyses of the *Thinker*, based on attention patterns, Logit Lens, and attention knockout, reinforce this picture. The model structurally processes audio tokens, but their direct causal influence on the final answer is weak. Blocking the direct audio channel has marginal effects, while disrupting the broader text-to-audio attentional structure in intermediate layers is more harmful. Overall, this thesis argues that evaluating Audio LLMs requires distinguishing structural multimodality from genuinely grounded listening, and provides tools to analyse not only whether a model is right, but how its answer is produced.

Key-words: Music Understanding, Audio Language Models, Explainable AI, Multimodal Interpretability.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	The black-box problem	3
1.3	Why Explainable AI	4
1.4	Benchmarks, Prompts, and the Evaluation of Listening	4
1.5	Model biases	5
1.6	Aim of the thesis	5
2	Background and Related Work	5
2.1	Background in Multimodal Machine Learning	5
2.2	Multimodal and Audio Large Language Models	6
2.3	Explainable AI	6
2.3.1	Explainability in multimodal models	7
2.3.2	Post-hoc methods	7
2.4	Shapley Values	8
2.4.1	Origin: Game Theory	8
2.4.2	Properties	8
2.4.3	SHAP framework	8
2.4.4	Permutation SHAP	8
2.5	MM-SHAP	9
2.6	LIME	9
2.7	DIME	10
2.7.1	Mathematical Formalisation	11
2.7.2	DIME limitations	12
2.8	Explainability in Multimodal Music Understanding	13
3	Methodology and Implementation	13
3.1	Pipeline Overview and Task Definition	13
3.2	Model Selection	14
3.3	Benchmarks and Datasets	15
3.4	Audio Representation	16
3.5	MM-SHAP Adaptation	17
3.6	Adapting DIME to the Audio Domain: audioDIME	18
3.6.1	Token-Based Scoring	19
3.6.2	Construction of UC and MI	19
3.6.3	Sampling Strategy and Computational Efficiency	20
3.6.4	Local Explanations with LIME	20
3.7	The Masking Problem	21
4	Analysis and Results Discussion	22
4.1	Computational Setup	22
4.2	Necessity and Sufficiency of audioDIME Explanations	23
4.3	Aggregate Analysis of Modal Behaviour	26
4.4	Investigating Textual Bias: Inside the Network	31
4.4.1	Attention Pattern	31
4.4.2	Logit Lens	33
4.4.3	Attention Knockout	35
5	Conclusions	38
6	Methodological Limitations	38
7	Future Work	39
A	Appendix A	42
B	Control Experiment with Silent Audio	43

1. Introduction

As early as 1950, Alan Turing formulated a question that would remain central to artificial intelligence: "*Can machines think?*". In *Computing Machinery and Intelligence*, Turing shifted the focus from an abstract definition of thought to the observable behaviour of a machine, proposing the well-known *imitation game* as an operational criterion for discussing intelligence. Reflecting on the nature of machine behaviour, he asked: "*May not machines carry out something which ought to be described as thinking but which is very different from what a man does?*"[28].

More than seventy years later, this question takes on a new form. Modern multimodal models no longer only manipulate textual symbols, but receive images, sounds, videos, and linguistic instructions, producing responses that are often coherent and convincing. However, the fact that a model provides a correct answer does not necessarily imply that it has used the relevant information in the intended way. In the case of Large Audio Language Models (LALMs), the question therefore becomes: *is the model genuinely listening to the audio signal, or is it producing a plausible response by exploiting textual knowledge, learned biases, or statistical shortcuts?* Answering this question is not possible by observing performance alone: it requires interpretative tools capable of analysing the internal decision-making process. This work aims to develop such tools.

1.1. Motivation

In recent years, the evolution of artificial intelligence models has led to a gradual move beyond purely unimodal systems. The most recent models no longer operate exclusively on text, images, or audio considered separately, but integrate heterogeneous sources of information within a single process of understanding and generation. This development reflects a fundamental characteristic of human experience: perception of the world is intrinsically multimodal, since events, objects, and meanings are interpreted by combining visual, acoustic, linguistic, and contextual signals [1].

Multimodal Machine Learning (MML) emerges precisely with the aim of building models capable of processing and relating information from multiple modalities. As highlighted by Baltrušaitis et al. [1], multimodality introduces specific challenges related to representation, alignment, fusion, and knowledge transfer across different sources. Similarly, Liang et al. [13] describe multimodal learning as a paradigm founded on three principles: modalities are *heterogeneous*, since they represent information through different spaces and structures; they are *connected*, as they often describe the same event or concept from different perspectives; and they are *interacting*, since their combination makes it possible to extract information that would not be accessible by considering them in isolation.

This scenario has gained further relevance with the emergence of Multimodal Large Language Models (MLLMs), in which the language model is used as a reasoning core capable of receiving and interpreting representations derived from multiple types of signals. In these systems, text is no longer the only source of information, but often becomes the medium through which the model formulates a response on the basis of heterogeneous inputs. In the musical domain, this architecture implies that the model must connect perceptual characteristics of the acoustic signal, such as timbre, rhythm, structure, pitch, and temporal relationships, to linguistic concepts that are often abstract. It is in this context that the need emerges for tools capable of going beyond performance evaluation and analysing the actual role of the different modalities in response generation.

1.2. The black-box problem

Although recent advances in Deep Neural Networks (DNNs) have led to increasingly advanced results in numerous perceptual and linguistic tasks, such models remain largely opaque systems from an interpretative point of view. In particular, modern deep learning-based models are characterised by high-dimensional architectures, millions or billions of parameters, and non-linear transformations distributed across multiple layers, making it difficult to understand explicitly how a given decision is produced.

This problem becomes even more evident as the scale of models increases. Large Language Models (LLMs) and, more recently, MLLMs exhibit increasingly sophisticated emergent capabilities, but at the cost of a growing loss of interpretability. As highlighted by Dang et al. [6], modern MLLMs operate as extremely complex black boxes, in which the internal decision-making process is difficult to access even for the developers of the model themselves. Consequently, observing only the final output is not sufficient to understand which information has actually been used during inference.

In the multimodal context, the black-box problem assumes an additional level of complexity. Unlike unimodal models, a multimodal system must process heterogeneous inputs, integrating them within shared representations. As Rodis et al. [21] state, this makes it difficult to distinguish the actual contribution of the individual modalities and to understand whether the model is genuinely exploiting cross-modal interactions or whether the

prediction depends predominantly on a dominant modality. They indeed emphasise how, in multimodal systems, explanation does not concern only which features are important, but also how information from different modalities is combined and mutually influences the model’s behaviour.

In the audio domain, these issues become even more critical. Audio is a continuous and temporal signal, characterised by overlapping sound events, perceptual variability, and dependencies distributed over time. As discussed by Çoban et al. [5], in audio-language models the connection between sound representation and textual reasoning may be incomplete, limiting the model’s ability to fully exploit the acoustic signal. This opacity is not merely a theoretical limitation: it makes it concretely difficult to validate whether an audio-language system actually processes the incoming sound content or is operating on different grounds. Addressing this problem requires analytical tools that go beyond performance measurement and make observable the process through which the different modalities contribute to the final response.

1.3. Why Explainable AI

Given the opacity of deep learning models, eXplainable AI (XAI) represents a fundamental tool for analysing, validating, and making artificial intelligence systems more reliable. The objective is not only to make predictions more understandable, but also to support model debugging, identify systematic errors, increase user trust, and detect possible biases in the decision-making process [4].

In multimodal models, however, explaining a prediction does not simply mean assigning importance to individual features. It is necessary to understand which modality contributes to the response, whether one modality dominates over the others, and how heterogeneous information is combined. Rodis et al. [21] indeed highlight that multimodal XAI must address specific problems related to attribution across modalities and to the explanation of inter-modal interactions.

This point is crucial because good performance does not necessarily guarantee that a model has learned genuine cross-modal interactions. For this reason, explainability methods for multimodal systems must not only identify which parts of the input are relevant, but also clarify whether the response depends on a single dominant modality or on the interaction between modalities.

The need for these tools emerges particularly clearly in Music Question Answering (MQA) benchmarks, where the correctness of a response does not guarantee genuine use of the audio signal. This empirical observation motivates the analysis developed in the following sections.

1.4. Benchmarks, Prompts, and the Evaluation of Listening

In MQA benchmarks, the model receives not only an acoustic signal, but also a question and a set of possible answers: the textual formulation of the prompt and of the distractors may therefore introduce sufficient cues to solve the task without a genuine perceptual analysis of the sound content.

This aspect has been highlighted by Zang et al. [34], who show that text-only models can achieve high performance on MQA benchmarks, reaching accuracies of up to 56.4% without any audio input, and that some LALMs maintain above-chance performance even when the real audio is replaced with Gaussian noise. In addition, Weck et al. [30] highlight how this *perception gap* suggests that many current benchmarks primarily measure the model’s textual reasoning ability, rather than genuine audio perception.

The formulation of the prompt therefore becomes a critical element of evaluation. In multimodal models, text is not simply a container for the request, but an informative modality that can strongly guide the final response. Çoban et al. [5] show how, in audio MLLMs, the connection between sound representation and textual reasoning may be incomplete, making it difficult to establish how much the response actually depends on the acoustic signal and how much on the linguistic content of the request.

These results indicate that MQA benchmarks must be designed to reduce textual shortcuts and dependencies on the structure of the answer options, prioritising questions that genuinely require audio perception. At the same time, they show that benchmark accuracy alone is not sufficient to evaluate listening: even when a model answers correctly, it remains necessary to verify whether the response is grounded in the acoustic signal or in textual cues.

This observation highlights a fundamental distinction between model performance and genuine multimodal understanding. A system may obtain correct answers without using acoustic information significantly in the decision-making process. Consequently, traditional metrics such as accuracy or benchmark scores must be complemented by interpretative analyses capable of measuring the actual contribution of the different modalities.

1.5. Model biases

Textual bias represents one of the most relevant forms of shortcut in MQA evaluation. When a benchmark can be partially solved from the question or the answer options alone, the model may appear to perform well without genuinely grounding its decision in acoustic information. For this reason, recent benchmarks attempt to reduce textual shortcuts by introducing questions written and validated by experts, more plausible distractors, and audio from controlled sources [30, 34].

A further dimension concerns cultural biases. LLMs are trained on large textual corpora that may overrepresent certain languages, geographical areas, social values, and cultural traditions. Tao et al. [27] show that different GPT models express cultural values closer to those of Anglophone countries and Protestant Europe, highlighting how models may incorporate a dominant cultural perspective. This aspect is relevant also in music, where genres, instruments, styles, and performance practices are strongly shaped by cultural and historical context. Consequently, a model may be more reliable on repertoires close to the dominant distribution of its training data, and less accurate on less represented musical traditions.

The issue of bias is directly connected to interpretability. In audio and multimodal systems, the opacity of models makes it difficult to establish whether a prediction derives from relevant acoustic information, textual cues, or spurious correlations. The analysis developed in this work therefore aims to distinguish between genuine use of the audio signal, dependence on text, and possible systematic biases.

1.6. Aim of the thesis

This thesis aims to develop interpretative tools for analysing the behaviour of multimodal audio-language models, with particular attention to how such models use audio and text when generating their responses. The work is organised around two independent and complementary analyses.

The first analysis extends MM-SHAP [19] to the context of Audio LLMs applied to music. MM-SHAP is used as a performance-agnostic measure: it does not evaluate whether the model’s answer is correct, but estimates the relative contribution of the available modalities to the generated response. This distinction is fundamental in systems that may answer correctly for reasons other than genuine use of the acoustic signal.

The second analysis introduces **audioDIME**, an adaptation of DIME [17] to the audio-musical domain. While DIME provides the theoretical framework for separating unimodal contributions from multimodal interactions, audioDIME reformulates this idea for generative audio-language models. In particular, it combines logit-based scoring, audio-text perturbations, source separation through Demucs, and onset-based segmentation in order to analyse not only how much audio and text contribute to the response, but also how their interaction influences the model’s decision.

Through this approach, the thesis aims to measure not only how much each modality contributes to the model’s final decision, but also whether audio and text are used independently or through genuine multimodal interaction. The proposed analyses are evaluated on HumMusQA [30], a controlled Music Question Answering benchmark chosen because it is designed to limit textual shortcuts and to require perceptual musical reasoning. The final objective is to provide empirical evidence on the actual behaviour of LALMs, distinguishing between genuine use of acoustic information, dependence on textual cues, and multimodal interaction.

The remainder of the thesis is organised as follows. Chapter 2 introduces the theoretical background and related work. Chapter 3 describes the proposed methodology, including the adaptation of MM-SHAP and the construction of audioDIME. Chapter 4 presents the experimental results and the internal analyses of the model. Finally, the thesis discusses the main conclusions, methodological limitations, and possible directions for future work.

2. Background and Related Work

This chapter introduces the theoretical and methodological background on which the thesis is based. It first presents the main concepts of Multimodal Machine Learning and Audio Large Language Models, then discusses the role of Explainable AI in multimodal systems. Finally, it reviews the post-hoc interpretability methods that are most relevant to this work, with particular attention to MM-SHAP, LIME, DIME, and their application to audio and music understanding.

2.1. Background in Multimodal Machine Learning

In order to build artificial intelligence systems capable of understanding the real world, it is not sufficient to analyse a single modality at a time, such as text, images, or audio; rather, it is necessary to integrate, align, and

combine information from different sources. Humans, indeed, perceive and interpret their environment in an intrinsically multimodal way, through visual, auditory, and linguistic signals. In this sense, Multimodal Machine Learning aims to model and exploit such heterogeneous information in order to improve the understanding and reasoning capabilities of artificial systems [1].

Building on the principles described by Liang et al. [13], Multimodal Learning can be understood as the problem of constructing representations that preserve the specificity of each modality while also making their relationships computationally usable. An effective multimodal model must therefore address three complementary requirements: representing heterogeneous signals in suitable latent spaces, aligning information that refers to the same event or concept, and fusing the modalities so that their interaction can support prediction or generation.

From an architectural perspective, the integration of modalities can take place according to different strategies. In *early fusion*, representations from different modalities are combined in the early stages of the model, allowing early interactions between signals to be learned. By contrast, in *late fusion*, each modality is processed separately and the information is aggregated only at the final decision-making level [1, 13]. Although these two strategies represent fundamental approaches, they are often limited in their ability to model complex relationships between modalities.

More recent Transformer-based models introduce more flexible mechanisms for multimodal integration. In particular, the attention paradigm, introduced by [29], makes it possible to dynamically weight the relevant information within a sequence. Extensions of this mechanism to the multimodal context have led to the development of *cross-attention*, in which one modality can dynamically select relevant information from another [33]. This approach makes it possible to model finer and more contextual interactions between modalities.

Representative examples of such architectures include models such as *ViLBERT* [14], which uses a dual-stream structure with co-attention mechanisms between text and images, and *VisualBERT* [12], which directly integrates visual and linguistic representations within a single Transformer. These models show how cross-modal attention can be used to learn semantic relationships between different modalities, for example between words and visual regions in vision-language tasks.

2.2. Multimodal and Audio Large Language Models

Multimodal Large Language Models extend the capabilities of traditional language models by integrating, in addition to text, modalities such as images, video, and audio. From this perspective, the language model is no longer used solely as a text generator, but as a reasoning core capable of operating on representations derived from heterogeneous sources. As described by Huang et al. [10], a typical pipeline for Large Multimodal Models includes an encoder specific to the input modality, an alignment module that projects these representations into a language-compatible space, and an LLM backbone responsible for generating the final response.

In the case of Large Audio Language Models, the acoustic input is first transformed into a latent representation through an audio encoder; this representation is then aligned with the textual space of the language model, enabling the system to respond to instructions, questions, or description requests related to the sound content. This architecture makes it possible to address tasks such as audio captioning, question answering, semantic classification, and reasoning over sound or musical events [5].

In the musical domain, these models play a particularly interesting role because they must connect perceptual characteristics of the audio signal, such as timbre, rhythm, structure, pitch, and temporal relationships, to linguistic concepts that are often abstract. The MUSE benchmark [3], for instance, was created precisely with the aim of evaluating musical perception abilities and auditory relational reasoning in multimodal models that integrate audio information, showing that musical understanding requires deeper capabilities than simple superficial classification.

However, the presence of an audio input does not automatically guarantee that the model uses it effectively. In Audio LLMs, the alignment between acoustic representations and textual reasoning may remain incomplete, and the final response can still be influenced by linguistic priors or task-specific shortcuts [5, 34]. This makes Audio LLMs a particularly relevant case for studying whether multimodal generation is genuinely grounded in the acoustic signal.

Consequently, Audio LLMs represent a natural extension of multimodal modelling towards the audio-musical domain, but also raise a fundamental interpretative question: whether the generated responses actually derive from the analysis of the acoustic signal, or from textual knowledge and biases learned during training.

2.3. Explainable AI

In the context of Explainable AI (XAI), it is essential to distinguish between *interpretability* and *explainability*, two concepts that are often confused but are conceptually distinct. Interpretability concerns the ability to

directly understand the internal functioning of a model, typically associated with intrinsically transparent systems. In the literature, models such as decision trees or linear regressions make it possible to explicitly trace the decision-making process, rendering the behaviour of the system immediately understandable [31].

By contrast, modern deep learning models are difficult to interpret, since they operate as black boxes, concealing the internal inferential process even from developers [31]. In these contexts, the role of explainability emerges, introducing *post-hoc* methods for explaining the model’s decisions after prediction, without requiring direct access to its internal structure.

Dang et al. [6] emphasise how this distinction is particularly relevant in next-generation language and multimodal models, where the high architectural complexity makes genuine interpretability impracticable, instead making explainability techniques necessary in order to analyse the model’s behaviour. Similarly, Rodis et al. [21] highlight that, in multimodal systems, explanation does not concern only the model itself, but also the contribution of the different input modalities, thereby further extending the concept of explainability.

Finally, Chakrabarty et al. [4] show how explainability may take different forms, for instance attributions, textual explanations, or visual explanations, all sharing the objective of making otherwise opaque decisions understandable. This confirms that, whereas interpretability implies a direct understanding of the model, explainability represents a set of tools required to explain complex systems retrospectively.

This distinction becomes particularly critical in multimodal models, where the need to explain interactions between different information sources introduces new challenges.

2.3.1 Explainability in multimodal models

In multimodal models, explainability assumes an even more complex role than in unimodal contexts, since the final prediction depends not only on the contributions of the individual modalities, but also on their interactions. In particular, one of the main problems is represented by *cross-modal interactions*, namely the way in which information from different modalities (for example audio and text) is combined during the inferential process. As highlighted by Hessel and Lee [8], distinguishing whether a model is actually exploiting such interactions or is instead relying predominantly on a single modality is a non-trivial problem, often not detectable through standard performance metrics.

A further critical aspect is *attribution ambiguity*, namely the difficulty of determining unambiguously which modality, or which combination of modalities, is responsible for the prediction [21]. In multimodal models, indeed, the presence of redundant or correlated signals across modalities may lead to misleading explanations, in which the importance attributed to a modality does not necessarily reflect its real causal contribution.

For this reason, recent surveys emphasise that explainability in MLLMs must go beyond techniques designed for unimodal models, explicitly accounting for both modality-specific contributions and inter-modal interactions [6, 21].

From a methodological perspective, Explainable AI is mainly articulated into two approaches: *intrinsic interpretability*, or transparency, and *post-hoc* methods. In line with the distinction introduced above, the former aims to design models whose internal decision-making process is explicit, whereas the latter analyses the behaviour of the model retrospectively, without modifying its structure, typically through input perturbation or feature attribution techniques.

2.3.2 Post-hoc methods

Within multimodal models, *post-hoc* methods currently represent the most widely used solution for analysing the behaviour of complex models, making it possible to explain predictions without modifying their internal structure.

A first approach consists in using input perturbation techniques, which estimate feature importance by observing the variation in the output as the input itself changes. As discussed by Morais et al. [18], these strategies make it possible to obtain an initial estimate of the relative contribution of the modalities, but they are limited when it comes to capturing complex multimodal interactions or non-linear effects. In particular, simple ablation operations, such as the complete removal of a modality, are not sufficient to describe the joint role of the information.

To overcome these limitations, more advanced methods based on local explanations have been developed. Among these, *LIME* provides local approximations of the model’s behaviour through interpretable models, while *SHAP* introduces a formulation based on Shapley values, guaranteeing desirable theoretical properties such as fairness and additivity. As highlighted by Chakrabarty et al. [4], these methods represent the state of the art for post-hoc interpretation, offering a good compromise between flexibility and theoretical robustness, while still presenting limitations related to computational cost and the stability of explanations.

Alongside approaches based on perturbations, gradient-based methods such as *Grad-CAM* [24] have also been employed in audio analysis, particularly when audio signals are represented as two-dimensional time-frequency representations (e.g., spectrograms or MFCCs) processed by convolutional architectures. Those methods make

it possible to visualise the most relevant regions of the input by exploiting gradient information internal to the model. However, while these methods effectively localise salient regions in the input, they are less suited for disentangling modality interactions in complex multimodal generative settings involving both audio and language.

2.4. Shapley Values

2.4.1 Origin: Game Theory

Shapley Values originate in cooperative game theory as a method for fairly distributing the overall payoff produced by a coalition of players [25]. The central idea is to assign each player a value proportional to their average marginal contribution, computed by considering all possible orders in which the players may enter the coalition. In the context of machine learning, this formulation is reinterpreted by considering input features as "players" and the model prediction as the value produced by the coalition [16].

Given a set of features N and a value function $v(S)$ associated with a subset $S \subseteq N$, the Shapley value of feature i is defined as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)].$$

The term $v(S \cup \{i\}) - v(S)$ represents the marginal contribution of feature i when it is added to the subset S , while the combinatorial coefficient weights this contribution with respect to all possible coalitions.

2.4.2 Properties

Shapley Values are particularly relevant for explainability because they satisfy theoretical properties that make them suitable for a fair distribution of contribution across features. The property of *symmetry* guarantees that two features that are equivalent with respect to the value function receive the same contribution. The property of *additivity* ensures that, if two games or models are combined, the contribution assigned to each feature is the sum of the contributions in the individual cases. Furthermore, the Shapley formulation implements a principle of *fairness*, since each feature is evaluated with respect to its average marginal contribution across all possible coalitions, avoiding arbitrarily privileging a specific order of inclusion [16, 25].

2.4.3 SHAP framework

The SHAP (*SHapley Additive exPlanations*) framework adapts Shapley Values to the interpretation of machine learning models, providing a unified formulation for attributing to each feature a share of the final prediction [16]. In this context, the SHAP value of a feature measures how much that feature contributes to shifting the model output with respect to a reference value, typically the expected value of the prediction. The sum of the SHAP contributions therefore reconstructs the difference between the model prediction and the baseline value [15].

This formulation is useful because it makes it possible to interpret even highly complex models through local attributions, that is, attributions referring to a specific prediction. However, the exact computation of Shapley Values requires evaluating all possible combinations of features, making it computationally prohibitive as the number of features grows.

2.4.4 Permutation SHAP

To make the computation of Shapley Values feasible, approximations are often used. *Permutation SHAP* estimates the average contributions of features by sampling different random orderings and measuring how the model output changes when a feature is progressively added to the coalition [15]. This makes it possible to avoid exhaustive computation over all subsets, reducing the computational cost compared with the exact formulation. However, even this approximation remains expensive, since it requires numerous forward passes of the model. In particular, the cost grows with the number of features and with the number of sampled permutations. For this reason, in multimodal and generative models, where features may correspond to textual tokens, audio segments, or visual regions, the definition of the features and the choice of masking become central aspects. This limitation is particularly relevant for methods based on Shapley Values, such as MM-SHAP, which extend this logic to the comparison between different modalities [18, 19].

2.5. MM-SHAP

MM-SHAP extends the SHAP framework to the multimodal context, with the aim of quantifying how much each modality contributes to a model’s prediction. Unlike metrics based exclusively on performance, such as accuracy, MM-SHAP is a *performance-agnostic* measure: it does not evaluate whether the model’s answer is correct, but rather the proportion in which the model uses the different available modalities [19]. This aspect is particularly relevant in multimodal models, where good performance does not necessarily imply a genuine joint use of the modalities.

The method is based on Shapley values, already introduced in the SHAP framework as a local attribution technique for estimating the contribution of individual features to the prediction [16]. In the multimodal case, features are grouped according to their modality, and the problem is reformulated by considering modalities as *coalitions* of features. In this way, the contribution of a modality is estimated as the average marginal contribution of the corresponding coalition with respect to all possible combinations of the other modalities.

The functioning of MM-SHAP therefore requires two fundamental steps: feature definition and multimodal masking. *Feature definition* establishes which units of the input are considered as elements to be perturbed, for example textual tokens, visual regions, or temporal segments of the audio. *Masking*, instead, consists in removing or replacing such features in order to measure the variation in the model output. By aggregating the Shapley values of the features belonging to the same modality, MM-SHAP obtains a modal contribution score, which can be normalised in order to directly compare the relative influence of the different modalities [19].

It is important to emphasise that the contributions estimated by MM-SHAP do not simply represent the absolute importance of a modality, but reflect the degree of dependence of the model on that modality in the context of the others. Consequently, a high contribution indicates that the model uses a modality significantly, but does not necessarily imply that such use is causal or semantically correct.

In the audio domain, Morais and Fuentes [18] adapt MM-SHAP to Audio LLMs in order to analyse whether models truly use acoustic information or rely predominantly on text. This adaptation shows how the framework can also be used beyond the vision-language domain, while still requiring specific choices for the definition of audio features and for signal masking.

Despite its usefulness, MM-SHAP presents some important limitations. First, the results depend strongly on the masking strategy adopted: a mask that is not neutral or not perceptually meaningful may alter the model’s behaviour and influence the interpretation. Second, the computational cost is high, since the method requires numerous forward passes over perturbed combinations of the input. Furthermore, the presence of correlations between modalities may make contribution attribution ambiguous, since redundant signals may be distributed across multiple modalities without reflecting their real causal role.

Finally, the interpretation of the scores is not always immediate: a high modal contribution indicates that a modality influences the output, but does not necessarily explain how the model internally processes that information. For this reason, MM-SHAP is particularly suitable for measuring the relative contribution of modalities, but it must be integrated with finer-grained analyses when the objective is to understand multimodal interactions.

2.6. LIME

LIME (*Local Interpretable Model-agnostic Explanations*) is a post-hoc explainability method proposed by Ribeiro et al. [20] with the aim of explaining individual predictions of black-box models in an interpretable and locally faithful manner. The idea underlying the method is that, even when the global behaviour of a complex model is not easily understandable, it can be locally approximated, in the neighbourhood of a specific input, by means of a simpler and interpretable model.

The functioning of LIME is based on the generation of perturbations of the original input. For each perturbation, the black-box model is queried again, and the variations in the output are used to estimate which components of the input have the greatest influence on the prediction. Starting from these perturbed examples, LIME trains an interpretable surrogate model, typically a linear one, assigning greater weight to the perturbations that are closer to the original instance. The coefficients of the surrogate model are then interpreted as the local importance of the features with respect to the prediction being analysed [20].

To make this interpretation possible, LIME introduces the concept of an *interpretable representation*. The original input x is in fact transformed into a simplified representation x' , consisting of components that are interpretable for humans [20]. In the textual domain, for example, such components may correspond to the presence or absence of words; in images, instead, to visual regions or superpixels. Perturbations are applied to this interpretable representation, making it possible to observe how the removal or modification of specific components influences the model’s prediction.

A central aspect of the framework is the distinction between local fidelity and interpretability. The surrogate model does not aim to reproduce the global behaviour of the black-box model, but only its behaviour in a local

region of the input space. In this way, LIME makes it possible to obtain understandable explanations even for highly complex models, while at the same time maintaining a certain degree of fidelity to the specific decision to be explained.

In this context, a fundamental trade-off emerges between the interpretability and the fidelity of the explanation. A surrogate model that is too simple may fail to correctly approximate the local behaviour of the black-box model, whereas an explanation that is too complex would risk losing readability and interpretative usefulness [20]. For this reason, LIME favours sparse explanations, based on a limited number of relevant features, so as to keep the explanation understandable for the end user.

The *model agnostic* nature of LIME represents one of its main strengths: the method does not require access to the model’s internal parameters, but only the possibility of observing its outputs. This makes it applicable to different domains and architectures, including text, images, and complex neural classifiers. In text, the features may be words or tokens; in images, superpixels or visual regions; more generally, the quality of the explanation depends strongly on how the features are defined and perturbed.

Despite its flexibility, LIME presents some limitations. The explanations depend on the perturbation strategy, on the definition of the interpretable features, and on the choice of local distance. Furthermore, since the method is based on sampling and local approximation, the explanations may be sensitive to the chosen parameters and do not necessarily reflect the global behaviour of the model. For this reason, LIME is particularly useful as a tool for local interpretation, but it requires care in defining the interpretable space and in reading the results.

A related issue concerns the granularity of textual features used in perturbation-based explanations. Although language models operate on tokens, token-level explanations are not always semantically transparent, especially when a tokenizer splits a meaningful word into multiple subword units. This problem is discussed by Kang et al. in AudioGenX [11], where the word “tiktoks” in the prompt “A clock tiktoks with music” is split into several fragments. The resulting attribution scores are distributed across subword pieces that are not all independently meaningful from a human perspective. This example shows that the interpretable representation used by LIME-style methods should not be confused with the model’s internal tokenisation: a technically valid token-level perturbation may still produce explanations that are difficult to interpret semantically.

This observation is particularly relevant for audio-language models, where textual prompts often contain words referring to sound events, instruments, or perceptual properties. If these words are fragmented into subword tokens, the resulting explanation may obscure the semantic role of the concept being analysed. For this reason, perturbation-based explanations require an explicit methodological choice about the level at which textual units are defined.

2.7. DIME

DIME (Disentangled Multimodal Explanations) [17] is a local interpretability framework designed to analyse black-box multimodal models through an explicit separation between unimodal contributions and multimodal interactions. The method arises from a well-known issue in the literature on multimodal models: good performance does not necessarily guarantee that the model is effectively exploiting integration between modalities. Indeed, a system may achieve high results by relying predominantly on a single modality or on shortcuts present in the data, without using genuine cross-modal interactions [8]. In this sense, DIME belongs to the context of multimodal explainability, where the objective is not only to identify which features are important, but also to distinguish whether the decision derives from a specific modality or from their combination.

The framework combines two complementary lines of research. On the one hand, it inherits from local explainability methods, such as LIME, the strategy of explaining the behaviour of a complex model in the neighbourhood of a specific input through an interpretable surrogate model [20]. On the other hand, it is inspired by the concept of disentanglement in representation learning, according to which a more interpretable representation should separate distinct explanatory factors instead of entangling different causes within the same representation [2, 9].

The concept of disentanglement originates from representation learning and refers to the ability to separate, within a representation, different explanatory factors of the data. Bengio et al. [2] highlight how entangled representations may hide the underlying structure of the data, making both learning and interpretation more difficult. In subsequent works, such as β -VAE [9], this principle was formalised through models that encourage the factorisation of latent variables, associating each latent dimension with a distinct semantic factor.

In the context of DIME, however, disentanglement takes on a different role. It is not used to learn new latent representations or to impose constraints during training, but rather as an interpretative tool for retrospectively decomposing the observable behaviour of the model. In particular, DIME distinguishes between what the model can infer by observing the modalities independently and what instead emerges exclusively from their combination [17]. This distinction is fundamental in multimodal models, where information from different modalities may be redundant, complementary, or interacting.

The main contribution of the framework therefore consists in modelling the multimodal prediction as composed

of two conceptually distinct components: *Unimodal Contributions* (UC), namely the information that each modality provides independently of the others, and *Multimodal Interactions* (MI), namely the information that emerges exclusively from the combination of modalities [17]. This distinction extends and makes more explanatory the idea introduced by EMAP (Empirical Multimodally-Additive function Projection) [8], which proposes to isolate the additive unimodal component in order to verify whether cross-modal interactions actually contribute to the prediction.

Formally, the model output can therefore be interpreted as the sum of a unimodal component and an interaction component. The former collects the additive contribution of the individual modalities, whereas the latter represents what cannot be explained by considering the modalities separately. This formulation directly reflects the problem highlighted by Hessel and Lee [8]: establishing whether a model is genuinely using cross-modal interactions is more complex than performance metrics alone would suggest.

A central aspect of the framework is that the separation between UC and MI has not only descriptive value, but also interpretative value. DIME makes it possible to distinguish whether a prediction is driven predominantly by unimodal signals, by multimodal interactions, or by a combination of the two, providing a useful tool for identifying undesirable behaviours such as dependence on unimodal biases or spurious correlations [17]. The value of disentanglement in DIME is therefore purely interpretative: it allows contributions of different natures to be analysed separately, clarifying whether the model is relying on a dominant modality or on a genuine integration between modalities.

In the context of DIME, LIME is used as a local explanation tool after the decomposition of the prediction into unimodal contributions and multimodal interactions [17]. Instead of directly explaining the overall multimodal prediction, the surrogate model is applied separately to the disentangled components of the decomposition. In this way, it becomes possible to distinguish the features relevant to unimodal contributions from those involved in multimodal interactions.

This setting overcomes a typical limitation of traditional attribution methods applied to multimodal inputs, where a single explanation may mix contributions of different natures. By explicitly separating unimodal contributions and interactions, DIME provides a finer-grained reading of the model’s decision-making process, useful for understanding not only which modalities are used, but also whether they are actually integrated. This approach is directly connected to the challenges of Multimodal Machine Learning, in which representation, alignment, and fusion between modalities constitute fundamental problems [1].

2.7.1 Mathematical Formalisation

From a formal perspective, DIME extends the paradigm of local explanations introduced by LIME by introducing an explicit decomposition of the multimodal prediction into unimodal contributions and multimodal interactions [17].

In the LIME framework, a local explanation is obtained by approximating the behaviour of the black-box model through an interpretable model defined in the local space of the input. Formally, given an instance x , LIME seeks an explanation $\xi(x)$ defined as:

$$\xi(x) = \arg \min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_x) + \Omega(g), \quad f : \mathcal{X} \rightarrow \mathbb{R}^C, \quad g : \{0, 1\}^d \rightarrow \mathbb{R}. \quad (1)$$

where f represents the black-box model to be explained, g is the interpretable surrogate model belonging to the family $\mathcal{G}$, $\mathcal{L}(f, g, \pi_x)$ measures how well the interpretable model locally approximates the behaviour of f , assigning greater weight to samples close to the original instance through the proximity function π_x , while $\Omega(g)$ represents a complexity term that favours simple and interpretable explanations [20]. Furthermore, $\mathcal{X}$ denotes the space of the original inputs, while C represents the dimensionality of the model output, corresponding to the number of logits or prediction classes. The interpretable model g operates instead on a binary interpretable space $\{0, 1\}^d$, where d is the number of interpretable features and each binary variable indicates the presence or absence of a feature in the perturbed sample.

DIME preserves LIME’s idea of local explanation, but introduces a further level of interpretation: instead of directly explaining the overall multimodal prediction, the framework first separates the model’s behaviour into unimodal and multimodal components [17].

Considering a multimodal model M that receives two modalities x_1 and x_2 as input, the output of the model can be expressed as:

$$V = M(x_1, x_2), \quad M : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}^C, \quad V \in \mathbb{R}^C. \quad (2)$$

where V represents the vector of logits produced by the model before the application of the softmax. The spaces $\mathcal{X}_1$ and $\mathcal{X}_2$ denote the input spaces associated with the two modalities, while the codomain $\mathbb{R}^C$ reflects the fact that the model produces a score for each of the C output classes.

The objective of DIME is to decompose the behaviour of the model in the form:

$$M(x_1, x_2) = g_1(x_1) + g_2(x_2) + g_{12}(x_1, x_2), \quad g_1 : \mathcal{X}_1 \rightarrow \mathbb{R}^C, \quad g_2 : \mathcal{X}_2 \rightarrow \mathbb{R}^C, \quad g_{12} : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}^C. \quad (3)$$

where $g_1(x_1)$ represents the unimodal contribution of the first modality, $g_2(x_2)$ represents the unimodal contribution of the second modality, and $g_{12}(x_1, x_2)$ captures the multimodal interactions between the two modalities. All three functions map to $\mathbb{R}^C$ since their contributions are defined in the same logit space as the original multimodal model M .

To isolate the unimodal contributions, DIME defines the *Unimodal Contributions* component as:

$$UC(M(x_1, x_2)) = \mathbb{E}_{x_1}[M(x_1, x_2)] + \mathbb{E}_{x_2}[M(x_1, x_2)] - \mathbb{E}_{x_1, x_2}[M(x_1, x_2)], \quad UC(M) : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}^C. \quad (4)$$

In this formulation, the expectations are computed by varying one modality at a time and observing the model’s response. The first two components respectively estimate the average contribution of the first and second modality, while the subtractive term avoids counting the component shared between the modalities twice [17]. The function $UC(M)$ preserves the same domain and codomain of the original multimodal model, since it still represents a multimodal prediction expressed in the logit space.

The *Multimodal Interactions* component is instead obtained as the residual of the decomposition:

$$MI(M(x_1, x_2)) = M(x_1, x_2) - UC(M(x_1, x_2)), \quad MI(M) : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}^C. \quad (5)$$

In this way, MI represents what cannot be explained by considering the modalities separately, namely the information that emerges exclusively from their combination. Since $MI(M)$ is obtained as the residual between two functions defined in $\mathbb{R}^C$, it preserves the same multimodal domain and output dimensionality.

Once the two components have been obtained, DIME applies LIME separately to the unimodal contributions and to the multimodal interactions, generating different local explanations for each modality [17]. In the two-modality case, the framework produces four explanations:

$$UC_i = LIME_i(UC_c(M)), \quad UC_c(M) : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}, \quad UC_i \in \mathbb{R}^{d_i}. \quad (6)$$

$$MI_i = LIME_i(MI_c(M)), \quad MI_c(M) : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}, \quad MI_i \in \mathbb{R}^{d_i}. \quad (7)$$

where $UC_c(M)$ and $MI_c(M)$ denote the scalar score associated with a specific target class c , extracted from the corresponding logit vector. Consequently, their codomain becomes $\mathbb{R}$. The vectors UC_i and MI_i belong instead to $\mathbb{R}^{d_i}$, where d_i is the number of interpretable features associated with modality i , since LIME produces a local attribution score for each interpretable feature.

This formulation makes it possible to explicitly distinguish the features that influence the prediction through independent contributions from those that emerge only through cross-modal integration, making DIME a framework particularly suited to the interpretative analysis of multimodal models.

2.7.2 DIME limitations

Although DIME offers an effective formulation for distinguishing between unimodal contributions and multimodal interactions, the method presents some important limitations. The first concerns the type of output required from the model: DIME is designed for black-box classification models that produce continuous values, such as the logits associated with classes. In [17], the authors highlight that the method works less well with models that return only discrete outputs, since input perturbations may not directly modify the final response and therefore may not produce informative variations on which to build the explanation.

A second limitation concerns scalability with respect to the number of modalities. In the original experiments, DIME is mainly applied to scenarios with two modalities, such as image and text. Extending it to three or more modalities significantly increases the computational cost, since estimating unimodal contributions and interactions requires evaluating an increasing number of combinations between inputs from different modalities [17]. Furthermore, as the number of modalities increases, the interpretation of the interactions also becomes more complex, including not only pairwise interactions but also higher-order interactions.

A further critical aspect concerns the application domain. Although DIME is formulated as a general method for multimodal models, the experimental validation in the original paper is concentrated mainly on image-text tasks, such as CLEVR and VQA. The authors themselves observe that the explanations generated through LIME are intuitive for images and text, whereas they may be more difficult to interpret in different modalities, for example temporal or vector signals [17]. This point is particularly relevant when the framework is considered for non-visual domains, where the definition of interpretable features and the visualisation of explanations require additional choices.

Finally, DIME inherits some limitations of local perturbation-based methods. The quality of the explanation depends on the choice of features, the perturbation strategy, and the number of samples used to estimate the model’s local behaviour. Consequently, although it provides a finer separation between unimodal contributions and multimodal interactions, DIME does not completely eliminate interpretative ambiguity and requires care in reading the results. These limitations motivate the need for specific adaptations when the method is applied to multimodal domains different from those originally studied.

2.8. Explainability in Multimodal Music Understanding

A first problem concerns segmentation. In order to explain an audio prediction, the signal must be divided into interpretable units, such as temporal intervals, frequency bands, or sound components. However, such units do not always correspond to perceptually meaningful elements for the listener. Perturbing an audio segment, for instance by silencing it or replacing it, may introduce acoustic artefacts and modify the perceptual content in an uncontrolled way, influencing the model’s response and therefore the quality of the explanation.

In this context, audioLIME [7] proposes an adaptation of LIME to the musical domain, introducing *listenable* explanations, that is, explanations that can be directly listened to by the user. Instead of arbitrarily perturbing portions of the signal, audioLIME uses components obtained through source separation and constructs perturbations by activating or deactivating such components. This approach makes the explanation closer to the perceptual experience of listening, since the user can acoustically verify which parts of the signal contribute to the prediction.

In the case of music, the complexity increases further because the relevant information may be distributed across multiple levels: timbre, rhythm, harmony, melody, structure, and temporal relationships. The MUSE benchmark [3] emphasises how musical understanding requires deeper perceptual and relational abilities than simple superficial classification, highlighting the difficulty of evaluating whether a model is genuinely processing meaningful musical properties. Consequently, explanations must also be able to represent not only "where" the model looks in the signal, but what type of musical information is being used.

MusicLIME [26] extends this perspective to the multimodal context, considering music understanding models that integrate audio and lyrics. The main contribution of the method is to show that explaining the modalities separately may be insufficient or misleading, because the prediction may depend on the interaction between acoustic content and textual content. MusicLIME therefore proposes a broader reading of explanations, in which feature importance is analysed by taking into account the joint contribution of the modalities.

These works show that explainability for audio cannot be limited to directly transferring methods developed for images or text. In the audio-musical domain, the definition of interpretable features, the choice of masking strategy, the perceptual fidelity of perturbations, and the readability of the resulting explanations become central methodological elements. For Audio LLMs, these challenges are further amplified by the multimodal nature of the task: explanations must account not only for the acoustic signal itself, but also for the way in which audio is combined with the textual prompt. This motivates the need for methods capable of analysing both modal contributions and audio-text interactions in a form that remains meaningful for music understanding [18].

3. Methodology and Implementation

This chapter describes the methodological and implementation choices adopted in this work. The proposed pipeline is designed to analyse the behaviour of a Large Audio Language Model in a multimodal Music Question Answering setting, with particular attention to the role played by audio, text, and their interaction in the generation of the final response. The chapter first introduces the task, the model, and the dataset, and then details the two independent interpretability analyses implemented in the pipeline: MM-SHAP and audioDIME. The full code for the methodology and experiments is available on GitHub: <https://github.com/fingenito/audioDIME>

3.1. Pipeline Overview and Task Definition

At the operational level, each evaluation sample is represented as a triplet composed of a musical audio clip, a textual question, and a set of candidate answers. The pipeline therefore treats the task as a controlled multimodal Music Question Answering problem, in which the model is queried with both the acoustic signal and the textual prompt, while the generated answer is analysed through the scores assigned to the candidate options.

Although the adopted model is generative in nature, the multiple-choice formulation provides a controlled interpretative target. Instead of analysing only the final generated answer, the pipeline operates on the logits associated with the candidate answer tokens before the final selection. In the case of multiple-choice QA, these

tokens typically correspond to the letters associated with the available options, such as A, B, C, and D. This formulation makes it possible to obtain a continuous signal on which perturbation-based interpretative methods can operate more reliably than in open-ended captioning settings, where multiple textual outputs may describe the same musical content.

For each sample, the audio is preprocessed by the model processor, while the textual prompt is tokenised and integrated with the acoustic input according to the format required by the multimodal model. Inference therefore produces a distribution of scores over the output tokens. The score associated with the target answer is used as the interpretative quantity for both MM-SHAP and audioDIME, as illustrated in Fig. 1.

The pipeline consists of two independent analyses designed to investigate complementary aspects of multimodal behaviour. The first analysis adapts MM-SHAP and estimates the relative contribution of audio and text with respect to the target response. Through controlled perturbations of the input, variations in the target score are measured in order to quantify the relative importance of the two modalities. This analysis provides an aggregated measure of modal contribution, but does not explicitly characterise the interactions between audio and text.

The second analysis, named *audioDIME*, adapts the DIME framework to the audio-musical domain. Unlike MM-SHAP, which provides a global estimate of modal contribution, audioDIME separates unimodal contributions from multimodal interactions, distinguishing what the model can infer from each modality individually from what emerges only from their combination. To make the audio domain compatible with this formulation, the pipeline integrates strategies inspired by audioLIME, based on source separation and interpretable temporal segmentation.

Both analyses rely on perturbations of the multimodal input. Each perturbation generates a modified version of the audio, the text, or both, which is then provided again to the model. Variations in the target score across perturbed inputs constitute the basis for estimating modal contributions, feature importance, and multimodal interactions.

Although MM-SHAP and audioDIME share some operational components, such as preprocessing, masking, logit-based scoring, and computational management, the two analyses remain methodologically independent. MM-SHAP provides a global measure of the relative contribution of the modalities, whereas audioDIME locally analyses the internal structure of the prediction by distinguishing between unimodal information and multimodal interaction.



Figure 1: General multimodal inference pipeline used in this work. The audio and the textual prompt are provided to Qwen2.5-Omni, which generates the response tokens associated with the multiple-choice task. The logits of the target tokens are subsequently used as the interpretative quantity in the MM-SHAP and audioDIME analyses.

3.2. Model Selection

The experiments were conducted using Qwen2.5-Omni-7B [32], an end-to-end multimodal model designed to process heterogeneous inputs, including text, images, audio, and video, and to generate textual or spoken responses in streaming mode. The choice of this model is motivated by the need for an audio-language system capable of receiving an acoustic signal and a textual prompt simultaneously, while at the same time keeping the generative behaviour accessible through the scores associated with the produced tokens.

From an architectural perspective, Qwen2.5-Omni adopts a *Thinker-Talker* structure: the *Thinker* module plays the role of the language model responsible for multimodal understanding and textual generation, while the *Talker* module uses the hidden representations produced by the Thinker to generate output audio tokens [32]. Since this work focuses on the analysis of textual responses and on the role of audio in inference, the methodological attention is placed mainly on the behaviour of the Thinker and on the logits associated with the generated textual tokens (Fig. 2).

A relevant aspect of Qwen2.5-Omni is the presence of audio and visual encoders capable of operating in a *block-wise* mode, designed to support multimodal streaming inputs [32]. The model also introduces TMRoPE (*Time-aligned Multimodal RoPE*), a variant of positional embedding designed to temporally align audio and video information [32]. Although the video component is not exploited in this work, these characteristics confirm that the model was designed to handle temporal multimodal signals and not simply to concatenate static representations.

Qwen2.5-Omni is particularly suitable for this context for three main reasons. First, the model is designed to integrate multiple modalities within a unified pipeline, making it coherent with the objective of analysing the relative role of audio and text in response generation. Second, its architecture makes it possible to work on the intermediate scores of the generated tokens, an essential aspect for the application of post-hoc methods such as MM-SHAP and DIME, which require a continuous signal on which to measure the effect of perturbations. Finally, Qwen2.5-Omni is designed for audio understanding and instruction-following scenarios, and is therefore compatible with Music Question Answering tasks and, more generally, with music understanding analyses. For these reasons, Qwen2.5-Omni was selected as the main model.

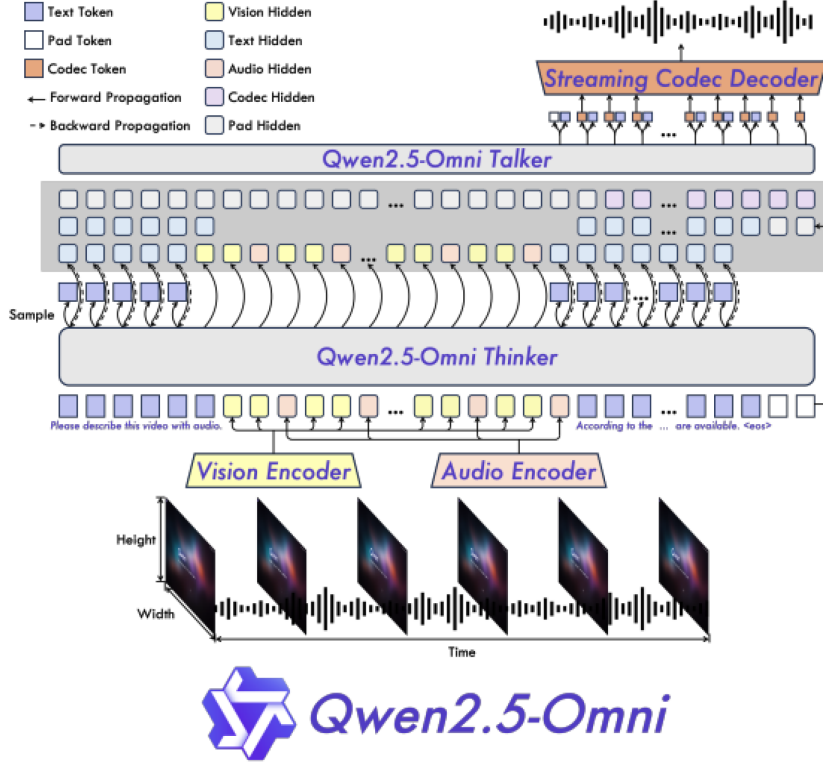


Figure 2: Thinker-Talker architecture of Qwen2.5-Omni [32]

3.3. Benchmarks and Datasets

The analysis focuses on HumMusQA [30], a Music Question Answering dataset constructed through expert annotation, designed to evaluate the ability of audio-language models to understand musical content and answer questions formulated in natural language.

HumMusQA was created with the aim of addressing a fundamental limitation of previous Audio Question Answering benchmarks: their strong dependence on text and the reduced need to genuinely use the audio during the inferential process. Recent studies have shown that many LALMs are able to answer correctly even when the audio signal is replaced with noise or silence, suggesting a strong reliance on linguistic priors rather than on genuine acoustic understanding.

To mitigate this problem, HumMusQA introduces questions designed to require greater perceptual grounding. The questions are written and validated by experts in music theory and cover different dimensions of musical listening, including: instrumentation, harmony, melody, metre and rhythm, performance, sound texture, and musical structure.

Each sample in the dataset consists of a musical audio clip, a natural-language question, a set of candidate answers, and a correct answer.

A particularly important aspect concerns the construction of the distractors. Unlike previous benchmarks, the incorrect answers are designed to be semantically plausible, reducing the possibility of eliminating incorrect options through superficial reasoning or linguistic shortcuts.

From a methodological point of view, HumMusQA is particularly suitable for the application of DIME. The multiple-choice task indeed makes it possible to directly associate the model’s decision with a continuous score relative to the candidate options, making it possible to apply the decomposition between unimodal contributions and multimodal interactions in a manner that is coherent with the formulation of the framework.

Furthermore, the variety of the questions makes it possible to observe very different scenarios. Some examples mainly require timbral or instrumental recognition, while others involve more theoretical or structural aspects of music. This makes HumMusQA a particularly informative context for studying the way in which audio and text are combined by the model.

Despite these improvements, the dataset does not completely eliminate the risk of linguistic shortcuts. Even in HumMusQA, some questions may be partially solvable through text alone, which is why the disentangled analysis proposed in this thesis assumes a particularly relevant role.

3.4. Audio Representation

Audio is the most critical component of the interpretative pipeline. Unlike text, which can be naturally divided into words or tokens, music audio is continuous, temporal, and composed of overlapping sound sources. To apply perturbation-based explainability methods, the waveform must therefore be transformed into interpretable and perturbable units.

In the proposed pipeline, the original audio signal is first separated into stems using Demucs [22], obtaining components such as vocals, drums, bass, and residual instruments. Each stem is then segmented using the same onset-based temporal grid, following a representation inspired by audioLIME [7]. As illustrated in Fig. 3, each audio feature corresponds to a source-segment pair rather than to a simple temporal window of the original waveform.

This choice is methodologically important because temporal segmentation alone may be poorly interpretable in the musical domain. Directly perturbing a window of the original waveform can modify several sound components simultaneously, making it difficult to determine whether an attribution is due to vocals, drums, bass, or other instruments. Preliminary source separation instead allows each perturbation to be associated with a more specific sound component.

The resulting audio features can therefore be represented as:

$$f_{s,t}, \quad f_{s,t} \in \mathcal{F}_a, \quad (8)$$

where $s \in \{1, \dots, S_a\}$ identifies the sound source, $t \in \{1, \dots, T_a\}$ denotes the temporal segment, and $\mathcal{F}_a$ represents the interpretable audio feature space. The final audio representation is therefore a source-time grid in which each cell corresponds to a binary or maskable feature.

This representation reduces overly generic attributions and provides explanations that are more coherent with musical perception.

The choice of temporal granularity remains a critical step. Segments that are too long risk mixing different musical events, reducing the localisation of the explanation. Segments that are too short increase the number of perturbable features, making the local surrogate model less stable and increasing the computational cost. In the present work, eight temporal segments of variable length, obtained through onset-based segmentation, were chosen as the standard.

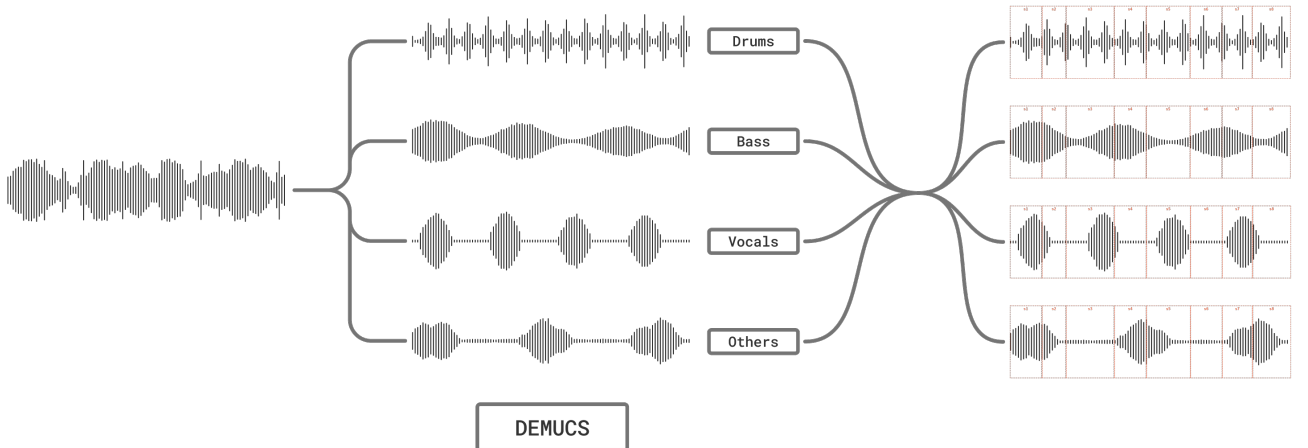


Figure 3: Source separation and feature extraction with Demucs and audioLIME. The original waveform is first separated into stems and each stem is then segmented using the same temporal grid, producing source-segment audio features.

3.5. MM-SHAP Adaptation

The first implemented analysis adapts MM-SHAP [18, 19] to the context of LALMs applied to Music Question Answering. The objective of the analysis is to estimate the relative contribution of the audio and text modalities with respect to the response generated by the model.

The implementation follows the original logic of MM-SHAP, but is adapted to a generative audio-language context based on Qwen2.5-Omni. In particular, the analysis does not only measure whether the model produces the correct answer, but how much that answer actually depends on the information coming from the audio and from the textual prompt.

For each HumMusQA sample, the model receives as input the original audio clip, an automatically constructed multiple-choice prompt, and four answer options.

A baseline response is initially generated using the complete audio-text input. The response produced by the model is then tokenised, and the generated tokens are used as the target of the SHAP analysis (Fig. 4).

As described in the general inference pipeline, the analysis is carried out on the logits associated with the tokens of the baseline response. This choice makes it possible to obtain a continuous signal compatible with the SHAP formulation and sufficiently stable to measure the effect of multimodal perturbations.

The multimodal features are constructed by concatenating audio features and textual features. The textual features correspond exclusively to the tokens of the question, while the prompt instructions and the multiple-choice options are kept fixed throughout the entire analysis. This choice prevents the perturbation from altering the format of the task or introducing artificial variations independent of the semantic content of the question. Unlike the onset-based segmentation used in audioDIME, here the waveform is divided into equal-length windows, so that the number of audio features matches the number of question tokens exactly:

$$N_A = N_T, \quad (9)$$

where N_A represents the number of audio segments and N_T the number of textual tokens in the question. Each audio feature therefore corresponds to a temporal window of the waveform.

The final input of MM-SHAP is therefore represented as a concatenation of multimodal features:

$$X = [A_1, \dots, A_{N_A}, T_1, \dots, T_{N_T}], \quad (10)$$

where A_i indicates an audio feature and T_j a textual feature.

The SHAP values are estimated through a permutation-based approximation. For each sample, a baseline response is first obtained from the complete, unmasked audio-text input. Then, for each sampled ordering of the multimodal features, the features are progressively added to an initially masked input following the selected order. At each step, the corresponding perturbed audio-text input is reconstructed and passed again through the model. The model produces a vector of logits over the vocabulary; from this vector, the scalar logit associated with the target token of the baseline response is extracted. The marginal variation of this scalar value after adding each feature is used to estimate the corresponding SHAP contribution. In the present work, $m = 30$ feature orderings are sampled for each analysed sample.

Since the model produces an output sequence, the SHAP values are computed separately for each generated token. The multimodal contributions are subsequently aggregated using the sum of the absolute values of the attributions:

$$\Phi_A = \sum_{i=1}^{d_A} |\phi_{A,i}|, \quad \Phi_T = \sum_{j=1}^{d_T} |\phi_{T,j}|, \quad \Phi_A, \Phi_T : \mathcal{X}_A \times \mathcal{X}_T \rightarrow \mathbb{R}_{\geq 0}. \quad (11)$$

The final MM-SHAP scores are then normalised as:

$$A\text{-SHAP} = \frac{\Phi_A}{\Phi_A + \Phi_T}, \quad T\text{-SHAP} = \frac{\Phi_T}{\Phi_A + \Phi_T}, \quad A\text{-SHAP}, T\text{-SHAP} : \mathcal{X}_A \times \mathcal{X}_T \rightarrow [0, 1]. \quad (12)$$

with:

$$A\text{-SHAP} + T\text{-SHAP} = 1. \quad (13)$$

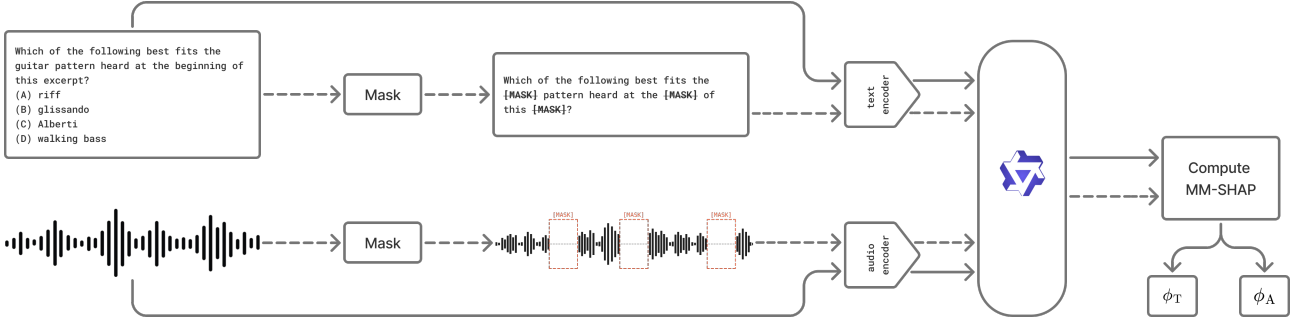


Figure 4: MM-SHAP pipeline [18].

3.6. Adapting DIME to the Audio Domain: audioDIME

The second implemented analysis, named *audioDIME*, adapts DIME to the audio-language context of Music Question Answering. The objective is not only to estimate how much audio and text influence the response, but to separate the part of the prediction that can be explained through additive unimodal effects from the part that emerges from the interaction between the two modalities.

Operationally, audioDIME follows a two-stage procedure. First, the score assigned by the model to the target response is decomposed into a unimodal component and a multimodal interaction component by evaluating the model on cross-combinations of audio signals and textual prompts from a background set. This stage determines how much of the target score can be attributed to audio and text considered additively, and how much remains as a residual audio-text interaction. Second, LIME is applied separately to these components and separately for each modality, in order to identify which audio source-segment features and which question words are responsible for the unimodal and interaction terms. The final output is therefore composed of four local explanations: UC_{audio} , UC_{text} , MI_{audio} , and MI_{text} .

In the present work, the multimodal model receives as input a musical audio signal x_a and a textual prompt x_t . The textual prompt is treated as a structured object composed of a fixed scaffold s , the question q , and the candidate answer options o :

$$x_t = (s, q, o). \quad (14)$$

Since Qwen2.5-Omni produces textual answers through an autoregressive language-model head, audioDIME is not applied to a single discrete class label, as in standard classification settings. Instead, it is applied to the pre-softmax logit associated with each target token of the generated response.

Given a target token y_k of the generated response, the value function used by audioDIME is defined by extracting the pre-softmax logit assigned to that token at generation step k . Let

$$\ell_k = F_\theta(x_a, x_t, y_{<k}), \quad \ell_k \in \mathbb{R}^{|\mathcal{V}|}, \quad (15)$$

where F_θ denotes the complete text-generation model up to the language-model head, $\mathcal{V}$ is the output vocabulary and $y_{<k} = (y_1, \dots, y_{k-1})$ denotes the sequence of target tokens preceding y_k . The vector ℓ_k contains one pre-softmax logit for each vocabulary token. The scalar value explained by audioDIME is then defined as:

$$m_k(x_a, x_t) = \ell_k[y_k], \quad m_k : \mathcal{X}_a \times \mathcal{X}_t \rightarrow \mathbb{R}, \quad (16)$$

where $\ell_k[y_k]$ indicates the entry of the logit vector corresponding to the target token y_k . In the autoregressive setting, this score is computed by conditioning the model on the audio x_a , the textual prompt x_t , and the previously generated target tokens $y_{<k}$. In this way, each target token is associated with a scalar pre-softmax score that can be used as a continuous value function. This preserves coherence with DIME, which operates on continuous outputs, while making the framework applicable to a text-generating language model.

Following the formulation of DIME [17], for each token y_k the prediction is decomposed into a component of *Unimodal Contributions* and a component of *Multimodal Interactions*. In the specific case of this thesis, the two modalities are audio and text, so the decomposition takes the form:

$$m_k(x_a, x_t) = UC_k(x_a, x_t) + MI_k(x_a, x_t), \quad UC_k, MI_k : \mathcal{X}_a \times \mathcal{X}_t \rightarrow \mathbb{R}. \quad (17)$$

The component UC_k represents the part of the score that can be explained through additive contributions from the two modalities, while MI_k represents the residual that cannot be explained by considering audio and text separately. Both components are scalar functions derived from the original token score $m_k(x_a, x_t)$ and are later explained independently through LIME. As a consequence, audioDIME does not produce a single attribution

map for the original prediction, but separate explanations for the additive unimodal component and for the residual multimodal interaction.

3.6.1 Token-Based Scoring

The use of token-level scores is the main adaptation required to apply DIME to a text-generating language model. In the original DIME formulation, the model output is typically a vector of class logits, and the explanation is built with respect to one selected class. In audioDIME, there is no fixed class output produced by the model. The generated response is instead a sequence of tokens, and each token position provides a separate pre-softmax logit that can be used as a continuous target.

For this reason, audioDIME treats each target token y_k as an independent explanation target. If the baseline response is composed of K tokens, the method produces K token-level decompositions and explanations. This formulation makes the generative output compatible with the DIME framework while preserving the autoregressive nature of the language model.

3.6.2 Construction of UC and MI

To estimate UC_k and MI_k , the pipeline constructs a background set composed of N audio-text pairs:

$$\mathcal{B} = \left\{ (x_a^{(i)}, x_t^{(i)}) \right\}_{i=1}^N, \quad x_a^{(i)} \in \mathcal{X}_a, \quad x_t^{(i)} \in \mathcal{X}_t. \quad (18)$$

Let r denote the index of the sample to be explained within the background set. In the implementation, the target sample is inserted as the first element of the background set, so that $r = 1$ and

$$(x_a^{(1)}, x_t^{(1)}) = (x_a, x_t). \quad (19)$$

The use of the first position is therefore only an indexing convention: the same decomposition can be written for any fixed index r .

For each target token y_k , a score matrix is constructed:

$$L_k[i, j] = m_k(x_a^{(i)}, x_t^{(j)}), \quad L_k \in \mathbb{R}^{N \times N}. \quad (20)$$

This matrix contains the scores obtained by evaluating the model on all cross-combinations between the audios and prompts in the background set. No part of the textual prompt is reordered or moved: each prompt is always provided to the model in its original form. The only operation is the recombination of complete audio samples with complete textual prompts from the background. Therefore, for a generic sample index r , the entry $L_k[r, r]$ corresponds to the original audio-text pair of that sample, row r keeps its audio fixed while varying the textual prompt, and column r keeps its textual prompt fixed while varying the audio.

Starting from L_k , the unimodal component for the sample indexed by r is estimated through the marginal means:

$$UC_k^{(r)} = \frac{1}{N} \sum_{j=1}^N L_k[r, j] + \frac{1}{N} \sum_{i=1}^N L_k[i, r] - \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N L_k[i, j]. \quad (21)$$

The first term keeps the audio of sample r fixed and averages over all textual prompts in the background set. The second term keeps the textual prompt of sample r fixed and averages over all background audios. The third term subtracts the global mean over all audio-text combinations, avoiding counting the shared baseline component twice.

The multimodal interaction component is then computed as the residual between the original score of sample r and its unimodal component:

$$MI_k^{(r)} = L_k[r, r] - UC_k^{(r)}. \quad (22)$$

This subtraction follows the residual definition of multimodal interaction in DIME. The term $L_k[r, r]$ is the complete score assigned to the target token when the original audio and textual prompt of sample r are provided together, while $UC_k^{(r)}$ estimates the part of this score explained by the additive contribution of the two modalities. The difference is therefore assigned to $MI_k^{(r)}$, which captures the part of the score that cannot be accounted for by audio and text considered separately.

Since the target sample is placed at index 1 in the implementation, the values used for the analysed instance are obtained as:

$$UC_k = UC_k^{(1)}, \quad MI_k = MI_k^{(1)}. \quad (23)$$

The numerical consistency of the decomposition is then checked through:

$$L_k[1, 1] \simeq UC_k + MI_k. \quad (24)$$

This check verifies that the original token score of the target sample is correctly reconstructed by the sum of its unimodal and interaction components. The equality is approximate because the computation is performed numerically and may be affected by floating-point precision. In practice, this control is useful for detecting implementation errors, such as incorrect indexing of the score matrix, mismatched audio-text pairs, or inconsistencies in the marginal sums.

3.6.3 Sampling Strategy and Computational Efficiency

The direct computation of DIME requires numerous forward passes, since for each token it is necessary to construct the matrix L_k and evaluate many local perturbations. To reduce the computational cost, the implementation follows the efficient strategy proposed by DIME [17], reusing the matrix L_k during the local explanation stage.

When the audio modality is perturbed, it is not necessary to reconstruct the entire matrix. For an audio perturbation $\tilde{x}_{a,s}$, it is sufficient to update the row corresponding to the target sample. More generally, if r denotes the index of the sample being explained, the updated scores are:

$$\tilde{L}_k[r, j] = m_k(\tilde{x}_{a,s}, x_t^{(j)}), \quad j = 1, \dots, N. \quad (25)$$

Similarly, when the text is perturbed, only the first column is updated:

$$\tilde{L}_k[i, r] = m_k(x_a^{(i)}, \tilde{x}_{t,s}), \quad i = 1, \dots, N. \quad (26)$$

From these updated rows or columns, UC_k and MI_k are recalculated for each perturbation. In this way, the pipeline avoids the complete reconstruction of L_k for every LIME sample. In the implementation, the update is not performed through an explicit copy of the entire matrix L_k , but through the efficient recomputation of only the marginal sums involved. This choice is mathematically equivalent to the row or column update required by the efficient DIME algorithm, but reduces the computational cost and the memory required.

Additional engineering optimisations, such as multi-GPU parallelisation, shared memory transfer of perturbed audio, and reuse of perturbation families across tokens, are described in the computational setup of the experiments.

3.6.4 Local Explanations with LIME

Once the scalar functions UC_k and MI_k have been obtained, LIME is applied separately to the two components and separately for each modality. In the bimodal case, the pipeline produces four local explanations: UC_{audio} , UC_{text} , MI_{audio} , and MI_{text} .

$$UC_{audio}, MI_{audio} \in \mathbb{R}^{d_a}, \quad UC_{text}, MI_{text} \in \mathbb{R}^{d_t}. \quad (27)$$

For each component and modality, LIME builds a local dataset composed of binary perturbation masks and the corresponding values of the function being explained. In the case of UC_k , the target values are the unimodal scores obtained after perturbation; in the case of MI_k , they are the corresponding interaction scores. A weighted linear surrogate model is then fitted on this local dataset, using the perturbation masks as interpretable inputs. Perturbations closer to the original sample are assigned larger weights through a kernel function, so that the surrogate approximates the behaviour of the model locally around the instance being explained. Ridge regularisation is used to stabilise the coefficients, and a preliminary feature selection step limits the fit to the most informative features, improving readability and reducing instability.

The audio representation follows a strategy inspired by audioLIME. Instead of directly perturbing temporal windows of the waveform, the signal is first separated into S_a stems and each stem is segmented into T_a temporal regions. The maximum number of audio features is therefore:

$$d_a = S_a \cdot T_a. \quad (28)$$

Each perturbation is encoded through a binary interpretable mask:

$$z_a \in \{0, 1\}^{d_a}, \quad (29)$$

where $z_{a,r} = 1$ indicates that audio feature r is kept, while $z_{a,r} = 0$ indicates that it is removed or attenuated in the perturbed reconstruction. This representation makes it possible to associate each attribution with a specific sound source and temporal interval, rather than with a generic waveform window.

For the audio modality, perturbations are therefore represented by binary masks applied to source-segment audio features. Each perturbation generates a new version of the audio and produces two target values, one for UC_k and one for MI_k . A local surrogate model is trained on these values. The resulting weights indicate, respectively, which audio features contribute unimodally and which are relevant for the interaction with text. On the textual side, perturbations are applied at word level only to the question, while keeping the prompt scaffolding and the answer options fixed. Textual perturbations are encoded as:

$$z_t \in \{0, 1\}^{d_t}, \quad (30)$$

where d_t is the number of perturbable words in the question. Masked words are replaced with the textual token [MASK]. This choice prevents the explanation from being dominated by perturbations of the candidate answers and focuses the analysis on the informative role of the question.

The selection of the most relevant features is limited to a predefined maximum number, fixed to 10 both for text and audio, so as to keep the explanations interpretable and reduce the instability of the surrogate model. For audio, the local fidelity value of the surrogate is also computed, expressed through R^2 , and used as an indicator of the quality of the local explanation.

In particular, features with a high weight in the MI_{audio} component indicate regions of the signal that become relevant mainly in relation to the specific textual question, while features with a high weight in UC_{audio} represent acoustic contributions that influence the response even independently of the linguistic content of the question.

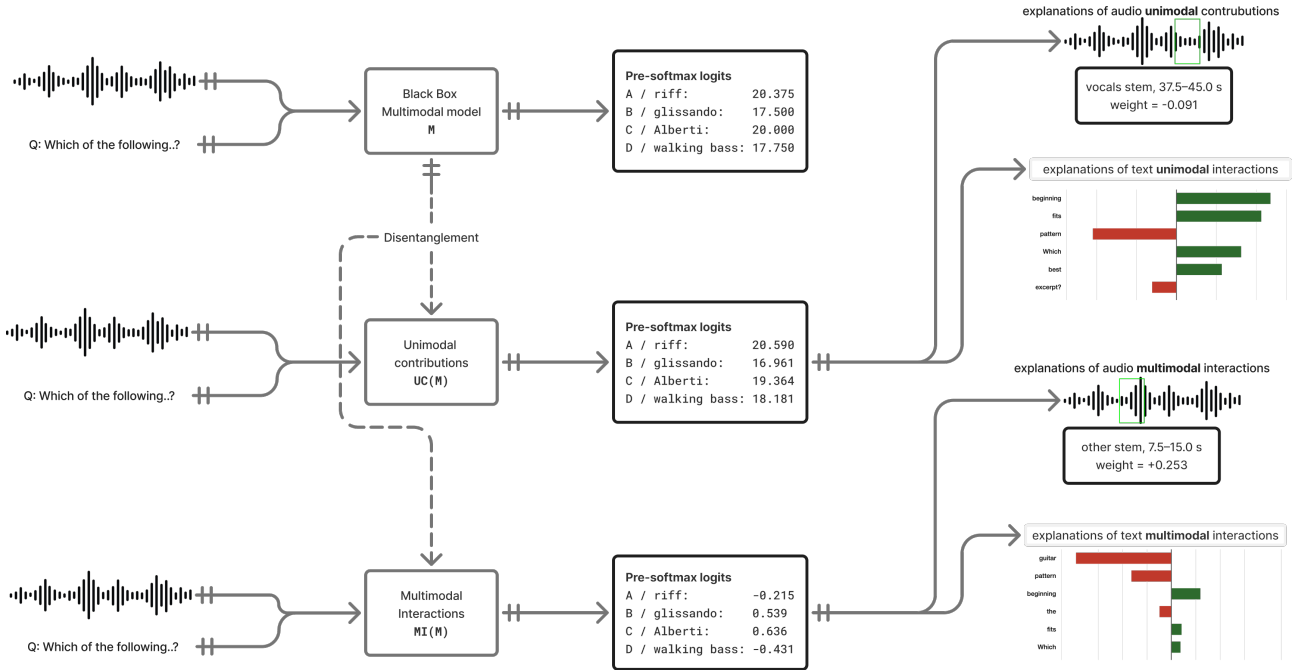


Figure 5: Example of the audioDIME pipeline applied to the first HumMusQA sample: the original multimodal prediction is decomposed into unimodal contributions $UC(M)$ and multimodal interactions $MI(M)$, with pre-softmax logits shown for each answer option. The right side reports the corresponding local explanations, highlighting the audio stem/time segment and question words most involved in each component.

3.7. The Masking Problem

Masking represents one of the most methodologically delicate aspects of the entire pipeline. Both MM-SHAP and audioDIME are based on input perturbations, but a perturbation is never completely neutral. If masking introduces strong artefacts or modifies the input distribution excessively, the model may react to the anomaly introduced by the perturbation rather than to the actual absence of the original information. For this reason, the masking strategy must be defined separately for audio and text, according to the structure of each modality. In the audio modality, the objective is to remove selected acoustic information while preserving a valid waveform. Different strategies were considered, including zeroing, replacement with noise, gradual attenuation, and fade-based masking. Zeroing is simple and reproducible, but may introduce discontinuities; noise replacement avoids silence but adds new acoustic energy; fade-based strategies are more perceptually plausible, but make feature removal less explicit.

In the final pipeline, MM-SHAP applies zeroing directly to selected temporal waveform windows. audioDIME instead applies binary masking to the audioLIME-inspired source-segment representation: the signal is first decomposed into stems and temporal segments, and components with mask value 0 are removed from the reconstructed waveform, while components with value 1 are retained. Thus, both analyses follow a removal-based masking logic, but at different representational levels: raw temporal windows for MM-SHAP and source-segment components for audioDIME.

In the textual modality, masking is applied only to the question q of the prompt, while the fixed scaffold s and the candidate answer options o are always preserved. Given the textual prompt

$$x_t = (s, q, o), \quad (31)$$

only the question component is converted into an interpretable representation. The question is segmented at word level:

$$q = (w_1, \dots, w_{d_t}), \quad (32)$$

where d_t is the number of perturbable words. A textual perturbation is represented by a binary mask

$$z_t \in \{0, 1\}^{d_t}, \quad (33)$$

where $z_{t,j} = 1$ indicates that the word w_j is kept, while $z_{t,j} = 0$ indicates that it is masked. Masked words are replaced with the literal token [MASK]. The perturbed textual input is therefore:

$$\tilde{x}_t(z_t) = (s, \tilde{q}(z_t), o), \quad (34)$$

where $\tilde{q}(z_t)$ denotes the masked version of the question.

This choice avoids perturbing the answer options, which would alter the decision space of the multiple-choice task and make the attribution less interpretable. By keeping the scaffold and the options fixed, the perturbation procedure isolates the effect of the linguistic content of the question. Word-level masking is preferred over token-level masking because it provides explanations that are more semantically readable: musically meaningful terms, such as instrument names or perceptual descriptors, are treated as interpretable units rather than as potentially fragmented subword tokens.

4. Analysis and Results Discussion

This chapter presents and discusses the experimental results of the thesis. The analysis first evaluates the causal faithfulness of audioDIME explanations, namely whether the features identified as relevant actually have a causal effect on the model’s decision when they are preserved or removed, through necessity and sufficiency tests. The chapter then analyses the aggregate modal behaviour of Qwen2.5-Omni by combining MM-SHAP and audioDIME metrics. Finally, the chapter moves inside the *Thinker* component, using attention patterns, Logit Lens projections, and attention knockout experiments to investigate the internal mechanisms associated with the observed textual bias.

4.1. Computational Setup

The execution of the experiments required careful management of computational resources, due to the high cost associated both with the inference of multimodal models and with the iterative application of explainability methods. In particular, the explainability techniques used involve a large number of forward passes, making optimisation strategies necessary.

The experiments were conducted on a system equipped with an Intel Xeon Gold 6246 CPU (48 cores @ 3.3 GHz) and 252 GiB of RAM, using 8 NVIDIA L4 GPUs, each with 24 GiB of memory. The use of the GPU was fundamental for handling high-dimensional audio inputs and for accelerating model inference. The batch size was set to 8, representing a compromise between computational efficiency and memory limitations, especially in the presence of audio inputs of variable length.

To reduce execution times, independent operations, such as the generation of perturbations and forward passes on distinct inputs, were parallelised through multiple processes, assigning separate workers to the available GPUs. However, some components of the adopted methods remain intrinsically sequential, limiting overall scalability.

Finally, an in-memory caching mechanism was introduced to reduce the number of redundant computations. In particular, the pipeline temporarily stores intermediate results such as the audio features extracted by the processor, the pre-processed inputs and, where possible, the common prefix of the textual context through reuse

of the model’s key-value cache. Furthermore, some audio representations are shared between processes through shared memory, avoiding redundant copies of the data across workers. This approach made it possible to reduce the cost of preprocessing and repeated forward passes, contributing to making the execution of the experiments more efficient.

Finally, the pipeline produces both aggregate measures of multimodal contributions and token-wise representations of the attributions, enabling local and global analyses of the model’s behaviour.

4.2. Necessity and Sufficiency of audioDIME Explanations

The design of this experiment is conceptually inspired by the factual and counterfactual evaluation logic adopted in AudioGenX [11]. In that work, factual reasoning identifies the information sufficient to reproduce the original prediction, whereas counterfactual reasoning identifies the information whose removal significantly changes that prediction. This thesis does not replicate the AudioGenX implementation, which relies on soft masks applied to cross-attention and on metrics defined in latent space or at the generative level. Instead, the same principle is adapted to the multiple-choice context of HumMusQA, defining necessity and sufficiency as probabilistic measures of the original target response, normalised with respect to the random-chance probability level $p_{\text{chance}} = 0.25$. The resulting metrics are therefore task-specific adaptations of the AudioGenX evaluation logic, not direct reproductions of its original metrics.

This experiment evaluates the causal faithfulness of the explanations produced by audioDIME: rather than measuring benchmark accuracy again, it verifies whether the features identified as relevant are actually decisive for the decision made by Qwen2.5-Omni. Since DIME decomposes the explanation into unimodal contributions UC_a , UC_t , and *Multimodal Interaction* (MI), the main analysis focuses on MI, which captures how audio and text jointly contribute to the response. The unimodal components are treated only as diagnostics, since evaluating them in isolation allows the model to compensate through the modality that remains intact and is therefore less informative with respect to genuinely multimodal behaviour.

For each sample, the model is first run on the complete audio-text input. The four logits associated with the multiple-choice options are converted into probabilities, and p_{orig} indicates the probability assigned to the model’s original target response, regardless of whether that response is correct.

It is important to clarify that, in this section, the terms *correct samples* and *incorrect samples* do not refer to the correctness of the explanation, but to the correctness of the model’s original prediction on the complete input. A sample is therefore defined as correct if the answer selected by Qwen2.5-Omni matches the dataset ground truth, whereas it is defined as incorrect if the answer selected by the model differs from the ground truth. In both cases, however, the analysis target remains the original response produced by the model. Consequently, for incorrect samples, audioDIME does not attempt to explain the correct answer of the dataset, but the decision actually made by the model. This distinction is essential, because the objective of the experiment is not to evaluate accuracy again, but to verify whether the features identified as relevant are causally linked to the observed decision, even when that decision is wrong.

Following the adapted factual/counterfactual protocol, progressive subsets of features ranked according to MI are constructed, with $k \in \{1, \dots, 16\}$, and their causal effect is measured through two complementary tests: sufficiency and necessity.

Sufficiency measures whether the top- k MI features are sufficient to preserve the model’s original decision. Operationally, only the selected features are kept, while all the others are masked. The model is then run again on the perturbed input, and the probability p_{suf} assigned to the same original target response is measured.

The metric is defined as:

$$\text{sufficiency} = \frac{p_{\text{suf}} - p_{\text{chance}}}{p_{\text{orig}} - p_{\text{chance}}}, \quad p_{\text{chance}} = 0.25. \quad (35)$$

Since the task is a four-option MCQA task, $p_{\text{chance}} = 0.25$ represents the chance-level probability of the analysed target option. The denominator ($p_{\text{orig}} - p_{\text{chance}}$) therefore represents the total informational advantage that the model possesses with respect to random chance when using the complete input, whereas the numerator ($p_{\text{suf}} - p_{\text{chance}}$) represents the part of that advantage that remains available when only the top- k features are used. Sufficiency therefore measures which fraction of the original useful information is preserved by the selected subset.

High values indicate that the selected subset preserves a large fraction of the model’s informational advantage. Values greater than 1 are possible when masking removes noisy or misleading features, increasing confidence with respect to the starting level.

Necessity instead measures whether the top- k MI features are necessary to maintain the model’s decision. In this case, the selected features are removed, while the rest of the input is kept unchanged. The model is then run again, obtaining the probability p_{nec} of the original target response.

The metric is defined as:

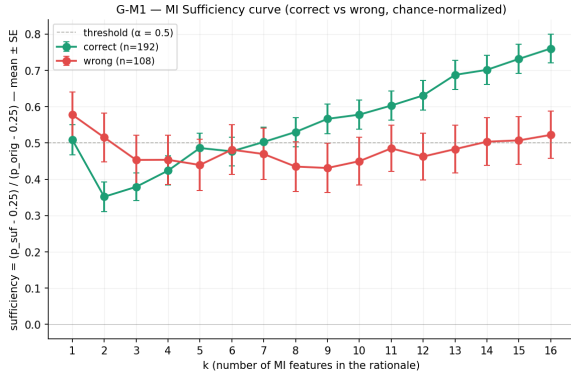
$$necessity = \frac{p_{\text{orig}} - p_{\text{nec}}}{p_{\text{orig}} - p_{\text{chance}}}. \quad (36)$$

Here too, the denominator represents the total informational advantage available with the complete input, whereas the numerator represents how much of that advantage is destroyed by removing the top- k features. Necessity therefore measures which fraction of the model’s useful information is lost by eliminating the selected subset.

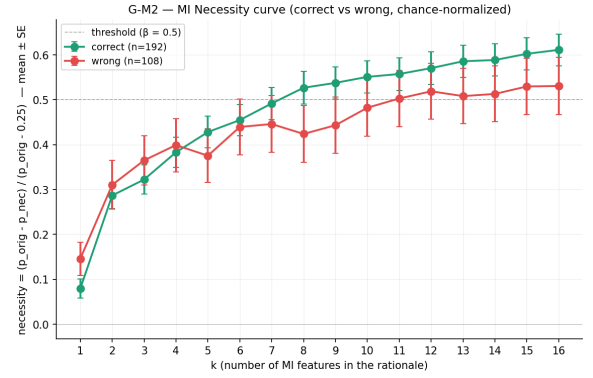
The main sufficiency and necessity curves are reported in Fig. 6. The comparison between correct and incorrect samples makes it possible to verify whether the causal faithfulness of the explanations changes as a function of the quality of the model’s original decision.

In both metrics, a value equal to 1 indicates that the selected subset entirely preserves or destroys the informational advantage with respect to random chance, whereas a value equal to 0 indicates behaviour equivalent to the random level. Negative values indicate that, after perturbation, the probability assigned to the target response falls below the random level. This suggests that the residual information is not simply insufficient, but becomes actively misleading with respect to the model’s original decision. The chosen operational threshold is equal to 0.5 for both sufficiency and necessity: this value can be interpreted as the point at which at least half of the original informational advantage is preserved or destroyed.

Samples for which the original probability is too close to random chance are excluded from the analysis, since the denominator $p_{\text{orig}} - p_{\text{chance}}$ would make the normalisation unstable. In practice, only sufficiently confident decisions are retained, requiring that the probability assigned to the original target response be at least $p_{\text{orig}} \geq 0.40$, above the random baseline $p_{\text{chance}} = 0.25$.



(a) Chance-normalised sufficiency of the top- k MI features as k increases, for correct samples ($n = 192$) and incorrect samples ($n = 108$), mean $\pm$ SE.



(b) Chance-normalised necessity of the top- k MI features as k increases, for correct and incorrect samples, mean $\pm$ SE. The dashed line indicates the operational threshold $\beta = 0.5$.

Figure 6: Causal faithfulness of the MI features as the number k of selected features varies ($k \in \{1, \dots, 16\}$), separating correct and incorrect samples. In correct samples, sufficiency progressively increases up to ≈ 0.76 at $k = 16$, with a drop at $k = 2$ due to the initial dominance of a single audio feature; necessity exceeds the 0.5 threshold around $k = 6$. In incorrect samples, both curves remain flatter, but necessity at small k is initially higher, indicating that wrong decisions depend on a small number of dominant features.

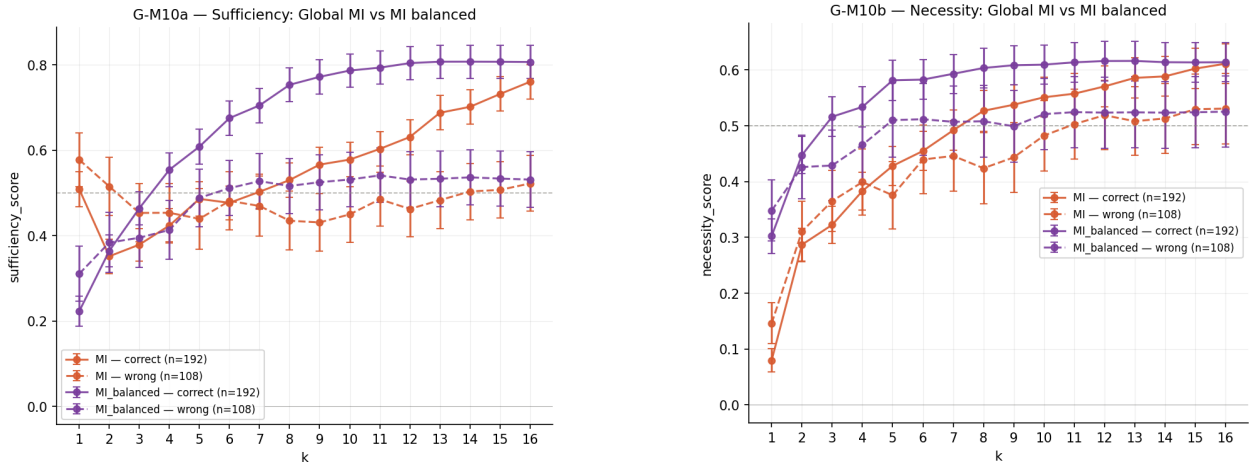
The curves in Fig. 6 show a clear difference between correct and incorrect samples. In particular, subfigure 6a shows that, for correct responses, sufficiency initially decreases, moving from approximately 0.51 at $k = 1$ to approximately 0.35 at $k = 2$, and then progressively increases up to approximately 0.75–0.76 at $k = 16$. This initial decrease suggests that the global MI ranking is often dominated by a single very strong feature, typically audio, while adding a second feature is not yet sufficient to reconstruct a genuinely multimodal context. As k increases, however, the explanation becomes more complete and is able to preserve a larger share of the original decision.

Necessity also increases regularly in correct samples, exceeding the 0.5 threshold around $k = 7$ and reaching approximately 0.61 at $k = 16$. As can be observed in subfigure 6b, incorrect samples instead show less coherent behaviour: sufficiency remains almost flat as k varies, while necessity increases more slowly and reaches the 0.5 threshold only for larger subsets. For small values of k , however, necessity is higher in incorrect samples, suggesting that wrong responses are often driven by a few misleading features. By contrast, correct decisions appear to depend on a broader and more redundant structure. Overall, this comparison supports the causal

relevance of the explanations produced by audioDIME, while also showing that explanation faithfulness is stronger when the underlying model decision is correct.

A second result concerns the composition of the selected MI features. The global MI ranking tends to select audio features first, especially for small values of k . This tendency, also observed in the top- k composition analysis, shows that audio features account for a dominant share in the first positions of the ranking, whereas textual features enter more stably only as the number of selected elements increases. Consequently, the explanations become clearly multimodal only for higher values of k . This aspect introduces a methodological limitation for sufficiency, because preserving a truly multimodal decision requires maintaining both sides of the audio-text interaction.

For this reason, a diagnostic variant, named MI balanced, was introduced, which selects the same number of audio and textual features. As shown in Fig. 7, the comparison between the global MI ranking and MI balanced highlights two distinct behaviours. In the case of sufficiency, reported in Fig. 7a, the balanced variant improves especially in correct samples and for small or intermediate values of k . In the case of necessity, shown in Fig. 7b, the advantage is instead more evident at small values of k and tends to decrease when the removed subset becomes large. Preserving a decision with few features therefore requires maintaining both modalities, whereas destroying it is possible even with an audio-dominated subset at high values of k .



(a) Sufficiency of the global MI ranking and of the MI balanced variant (equal number of audio and textual features for each k), for correct and incorrect samples.

(b) Necessity of the global MI ranking and of the MI balanced variant, for correct and incorrect samples.

Figure 7: Comparison between the global MI ranking and the MI balanced variant, which selects the same number of audio and textual features. In sufficiency, the balanced variant clearly outperforms the global ranking in correct samples (≈ 0.81 vs ≈ 0.76 at $k = 16$), confirming that preserving a multimodal decision requires features from both modalities. In necessity, the advantage of balancing is already evident at small k and becomes weaker for larger subsets, where audio alone is sufficient to destabilise the decision.

From a methodological point of view, global MI and MI balanced should not be considered conflicting alternatives. The global ranking shows which interactions spontaneously emerge as strongest in the model, whereas MI balanced is better suited to testing the causal sufficiency of a genuinely multimodal explanation. In this sense, MI balanced does not replace the main analysis, but clarifies one of its operational limitations: a free ranking may be correct as an ordering of importance, but not necessarily optimal for constructing a minimal subset capable of preserving the decision.

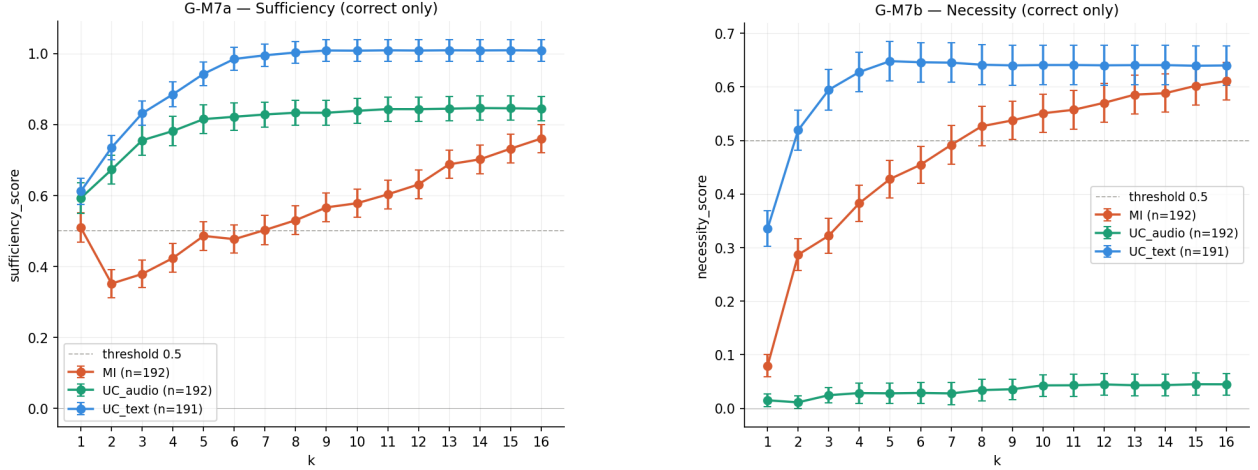
The secondary results, although not reported as main figures, confirm this interpretation. The distribution of the minimum value of k shows that preserving the decision is generally easier than destroying it: in many cases, few features are sufficient to maintain a relevant share of the model’s confidence, whereas removing the relevant features often requires larger subsets. Moreover, a non-negligible share of samples does not reach the necessity threshold within $k = 16$, indicating that the model can compensate for the removal of the selected features through alternative information present in the input. These results reinforce the idea that the model’s decision is often redundant, especially in correct samples.

Finally, the diagnostic comparison between UC_{audio} , UC_{text} , and MI clarifies the modal structure of the model’s decision (Fig. 8). In the case of sufficiency, reported in Fig. 8a, UC_{text} reaches very high values, indicating that the most relevant words in the question are often sufficient to preserve the model’s response. In the case of

necessity, shown in Fig. 8b, the removal of the same textual features instead substantially weakens the original response. This confirms that the textual component is not only sufficient, but also necessary for the model’s decision.

UC_{audio} , by contrast, exhibits an asymmetric behaviour: it reaches relatively high sufficiency, but remains almost non-necessary. In other words, audio information can support the decision when it is preserved, but its removal produces only a very limited loss in the probability of the original response. This indicates that unimodal audio features are often replaceable by textual signals.

The MI component occupies an intermediate position. It is less sufficient than UC_{text} , but more necessary than UC_{audio} , suggesting that audio becomes causally relevant especially when it participates in an interaction with the textual question. This is the central result of the experiment: Qwen2.5-Omni does not ignore audio, but the strongest causal role of audio emerges when it is mediated by text, rather than when it is considered as an autonomous unimodal signal.



(a) Chance-normalised sufficiency of UC_{audio} , UC_{text} , and MI in correct samples. UC_{text} reaches saturation (≈ 1.0) already around $k = 6$, whereas UC_{audio} stabilises at ≈ 0.83 and MI increases more slowly up to ≈ 0.76 .

(b) Chance-normalised necessity of UC_{audio} , UC_{text} , and MI in correct samples. UC_{audio} remains close to zero for all values of k (≈ 0.03), indicating that unimodal audio features are almost entirely replaceable by the textual component.

Figure 8: Modal diagnosis of the model’s decision in correct samples: sufficiency (a) and necessity (b) chance-normalised for UC_{audio} , UC_{text} , and MI. UC_{text} is the dominant component in both metrics (suf ≈ 1.0 , nec ≈ 0.65), confirming the model’s text-first behaviour. UC_{audio} shows a clear causal asymmetry: moderate sufficiency (≈ 0.83) but almost null necessity (≈ 0.03), indicating that unimodal audio features are replaceable by text. MI occupies an intermediate position (suf ≈ 0.76 , nec ≈ 0.62), suggesting that audio acquires causal relevance mainly in interaction with the textual component.

Overall, the experiment leads to three main conclusions. First, audioDIME produces explanations with measurable causal relevance: the features identified by the MI component can preserve and weaken the original decision. Second, this faithfulness is stronger for correct responses, indicating that correct decisions are associated with more coherent and redundant multimodal rationales. Third, the model shows a clear text-first behaviour: textual features are highly sufficient and necessary, whereas audio features are often replaceable, unless they enter the multimodal interaction component. The main role of audio is therefore not purely unimodal, but interactional.

4.3. Aggregate Analysis of Modal Behaviour

The objective of the experiment is to characterise, on an aggregate scale, the modal behaviour of Qwen2.5-Omni on the HumMusQA dataset. Unlike the causal faithfulness experiments, this analysis does not directly perturb the features in order to verify their effect on the decision, but combines the information produced by the two main analysis methodologies: MM-SHAP and audioDIME.

For each sample, A -SHAP and T -SHAP are therefore extracted, together with the audioDIME components UC_a , UC_t , MI_a , and MI_t . In this section, UC_a , UC_t , MI_a , and MI_t denote the local attribution vectors produced by LIME for the audio and textual features, and not the scalar scores of the token-level decomposition. In particular, MI_a contains the attributions associated with the audio source-segment features, whereas MI_t

contains the attributions associated with the perturbable words of the question. The same considerations apply to UC_a and UC_t .

Starting from these attribution vectors, the multimodal interaction ratio is defined:

$$\text{MI-ratio} = \frac{\|MI^{attr}\|_1}{\|UC^{attr}\|_1 + \|MI^{attr}\|_1}, \quad (37)$$

where

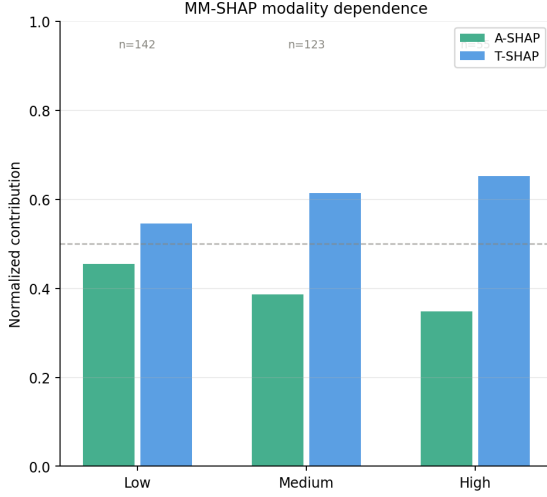
$$\|MI^{attr}\|_1 = \|MI_a\|_1 + \|MI_t\|_1 = \sum_i |MI_{a,i}| + \sum_j |MI_{t,j}|, \quad (38)$$

and, analogously,

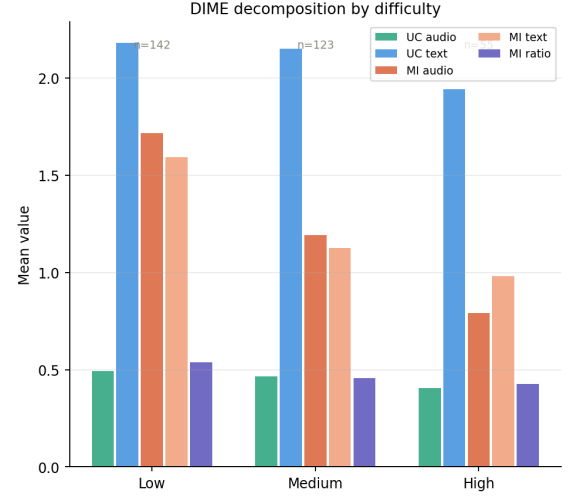
$$\|UC^{attr}\|_1 = \|UC_a\|_1 + \|UC_t\|_1. \quad (39)$$

The subscript 1 therefore denotes the L_1 norm, computed as the sum of the absolute values of the attributions. This aggregation is used because audioDIME produces signed feature-level contributions: using absolute values prevents positive and negative effects from cancelling each other out, making it possible to measure the overall intensity of the interaction component. The MI-ratio therefore measures the relative share of attributional intensity assigned to the audio-text interaction with respect to the total contribution estimated by audioDIME. It is a normalised derived quantity, not directly comparable with the absolute values of UC and MI, but provides a compact indicator of the degree of multimodal behaviour.

The analysis of these quantities over the full set of samples reveals a coherent text-first pattern, whose structure varies with difficulty, category, and response correctness. In almost all conditions, T -SHAP is greater than A -SHAP, indicating that the model’s global decision is supported more strongly by the textual prompt and the answer options than by the audio signal. This trend is particularly evident as difficulty increases (Fig. 9). In particular, subfigure 9a shows that, in *low* questions, the mean audio contribution is approximately 0.45, while the textual contribution is approximately 0.55; in *medium* questions, A -SHAP decreases to approximately 0.39, with T -SHAP around 0.61; in *high* questions, the audio contribution decreases further to approximately 0.35, while the textual contribution reaches approximately 0.65. Subfigure 9b instead reports the aggregate trend of the audioDIME components as difficulty varies, showing that the unimodal contribution of text remains dominant even when the task conditions change. This suggests that, when the task becomes more difficult, the model tends to compensate for perceptual uncertainty by relying more strongly on linguistic information, answer options, and prior knowledge.



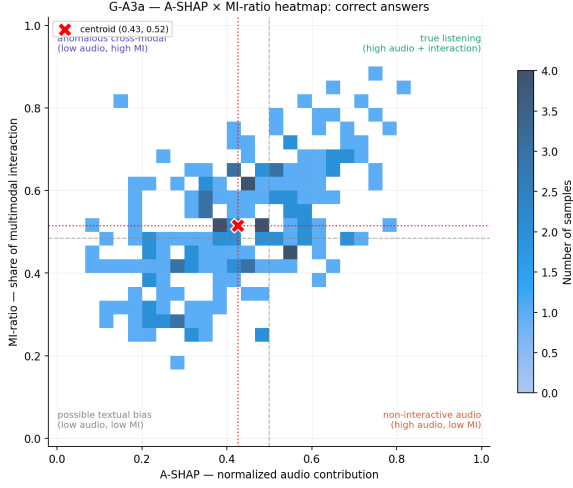
(a) Relative A -SHAP and T -SHAP contributions by difficulty level (*low* $n = 142$, *medium* $n = 123$, *high* $n \approx 55$). A -SHAP decreases from ≈ 0.46 to ≈ 0.35 as difficulty increases; the dashed line indicates audio-text parity at 0.5.



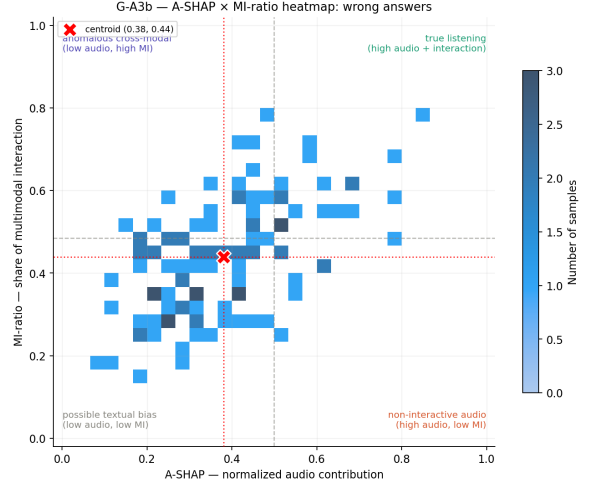
(b) Mean audioDIME components by difficulty level (mean L_1 norms of the attributions, plus MI-ratio). UC_{text} remains dominant (≈ 2.2) at all levels, while MI_{audio} and MI-ratio progressively decrease with difficulty.

Figure 9: XAI profile by difficulty level: MM-SHAP contributions (a) and audioDIME components (b). As difficulty increases, A -SHAP decreases from ≈ 0.46 (*low*) to ≈ 0.35 (*high*) and T -SHAP correspondingly rises from ≈ 0.54 to ≈ 0.65 ; the multimodal interaction components (MI_{audio} , MI_{text} , MI-ratio) show the same decreasing trend. This suggests that, when the task becomes more difficult, the model compensates for perceptual uncertainty by relying more strongly on linguistic information.

The audioDIME decomposition confirms this interpretation, but introduces an important distinction. The UC_t component is the highest in almost all conditions, showing that the unimodal contribution of text constitutes a dominant part of the explanation. However, UC_t should not be interpreted as an indicator of correctness: it remains high in both correct and incorrect samples. The most relevant difference between correct and incorrect responses instead emerges in the multimodal interaction components. In correct samples, MI_a , MI_t , and MI-ratio are higher than in incorrect samples, whereas the difference in T -SHAP is more limited. In other words, text is used intensively in both groups, but what distinguishes correct responses is greater audio-text integration.



(a) Joint distribution of A-SHAP and MI-ratio for samples with a correct response. The centroid is (0.43, 0.52); the four quadrants identify four modal regimes (labels in the figure).

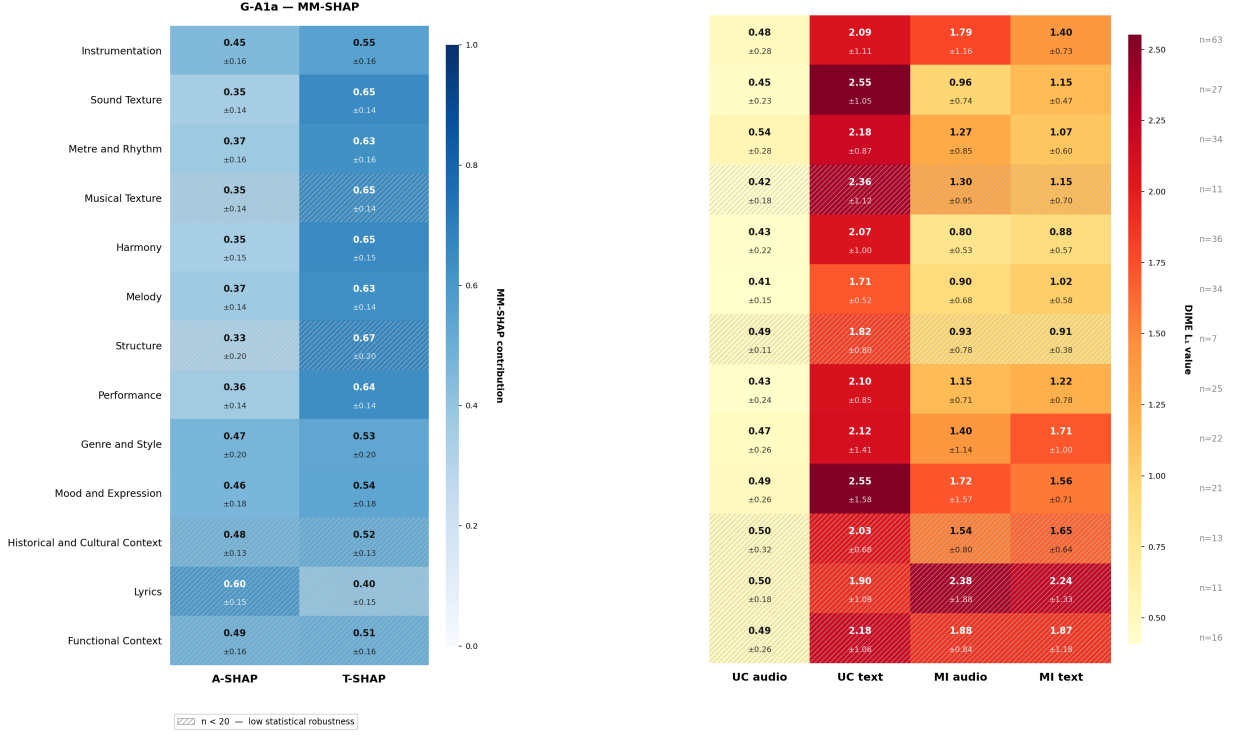


(b) Joint distribution of A-SHAP and MI-ratio for samples with an incorrect response. The centroid is (0.38, 0.44), shifted towards lower values compared with correct samples.

Figure 10: Joint distribution of A-SHAP (normalised audio contribution) and MI-ratio (share of multimodal interaction) for correct (a) and incorrect (b) samples. Correct samples cluster around the centroid (0.43, 0.52), compared with (0.38, 0.44) for incorrect ones, indicating that correct responses are associated on average with greater audio use and a higher share of multimodal interaction. Both distributions are broad; the centroids describe an aggregate tendency, not a uniform behaviour across individual samples.

The joint distribution of A-SHAP and MI-ratio confirms this picture (Fig. 10). Subfigure 10a shows the distribution of correct samples, while subfigure 10b reports that of incorrect samples. Correct samples cluster around the centroid (0.43, 0.52), compared with (0.38, 0.44) for incorrect ones, indicating that correct responses are associated on average with a greater audio contribution and a higher share of multimodal interaction. This result is central: accuracy does not depend simply on the use of text, but on the model’s ability to use it as a guide for interpreting audio, rather than as an autonomous shortcut. At the same time, the two heatmaps show broad, non-concentrated distributions; the centroids therefore describe an aggregate tendency, not a uniform behaviour across all samples.

The category-level analysis adds a more specifically musical interpretation, but must be treated with caution (Fig. 11). Subfigure 11a shows the relative audio/text contributions estimated by MM-SHAP, whereas subfigure 11b reports the unimodal and interaction components estimated by audioDIME. Some categories, including *Lyrics*, *Functional Context*, *Mood and Expression*, *Historical and Cultural Context*, and *Genre and Style*, present relatively higher values of A-SHAP or multimodal interaction. The case of *Lyrics* is the most evident: A-SHAP reaches approximately 0.60, while T-SHAP decreases to approximately 0.40, suggesting a more direct dependence on vocal audio. However, the same category also shows a very high standard deviation for MI_a (2.38 ± 1.88), indicating strong internal variability. Category means should therefore not be read as uniform behaviours, but as aggregate trends that may hide relevant differences across individual questions.



(a) Heatmap of A-SHAP and T-SHAP contributions by musical category (mean $\pm$ standard deviation). Categories with $n < 20$ samples are hatched (low statistical robustness). *Lyrics* reaches the highest A-SHAP value (≈ 0.60).

(b) Heatmap of audioDIME components by musical category (mean $\pm$ standard deviation, column-wise normalisation). UC_{text} is dominant in all categories; *Lyrics* presents the highest values of MI_{audio} (≈ 2.38) and MI_{text} (≈ 2.24), with strong internal variability.

Figure 11: Mean XAI profiles by musical category: normalised MM-SHAP contributions (a) and audioDIME components (b), column-wise normalised. *Lyrics* is the category with the highest audio contribution ($A\text{-SHAP} \approx 0.60$) and the highest multimodal interaction values ($MI_{audio} \approx 2.38 \pm 1.88$); categories such as *Functional Context*, *Mood and Expression*, and *Historical and Cultural Context* show audio contributions moderately above average. The column-wise normalisation in (b) reflects variation within each metric; categories with $n < 20$ samples (hatched in (a)) have low statistical robustness and should be interpreted with caution.

The secondary results confirm and refine this interpretation. The difficulty heatmaps, although not reported as main figures, show that the relationship between audio contribution and MI-ratio remains broad even within the same difficulty level: there are therefore difficult samples with a relatively high share of audio and multimodal interaction, as well as easier samples solved with strong textual prevalence. This confirms that difficulty describes an average tendency, but does not rigidly determine the model’s modal behaviour. Similarly, category differences indicate that the type of musical information required by the question influences the balance between audio, text, and interaction, without eliminating the general text-first pattern.

Three main limitations must be considered. First, the MM-SHAP and audioDIME scales are not homogeneous: A-SHAP and T-SHAP are modality-normalised contributions, whereas the audioDIME components are aggregated through the L_1 norm, which removes the sign and measures global intensity but not the direction of individual effects. Second, the category heatmap is column-wise normalised, so colours reflect variation within each metric rather than direct comparisons between different metrics. Finally, difficulty levels are unbalanced: *low* and *medium* contain 142 and 123 samples respectively, whereas *high* contains approximately 55. Consequently, the results for the most difficult questions should be considered less statistically robust than those for the other levels.

Overall, this experiment shows that Qwen2.5-Omni operates on HumMusQA according to a predominantly text-first dynamic. Text structures the decision in almost all conditions, but response correctness is associated with greater audio-text integration. The Multimodal Interaction components are indeed the most informative discriminators between correct and incorrect responses, whereas the unimodal contribution of text remains high in both cases. As difficulty increases, the model reduces the audio contribution estimated by MM-SHAP and tends to rely more strongly on text, suggesting greater dependence on linguistic priors when acoustic information

is harder to use. The experiment therefore provides a quantitative characterisation of the model’s modal behaviour: Qwen2.5-Omni does not listen uniformly, but integrates audio most effectively when it succeeds in constructing a genuine multimodal interaction with the textual prompt.

4.4. Investigating Textual Bias: Inside the Network

After observing, through MM-SHAP and audioDIME, a predominantly text-first behaviour, the analysis moves from the post-hoc level to the internal functioning of the model. The objective is to verify whether textual bias also emerges in the computational mechanisms of the *Thinker* of Qwen2.5-Omni-7B. The experimental design follows the mechanistic interpretability setting proposed by Selvakumar et al. [23], adapting it to the audio-textual MQA scenario considered in this work.

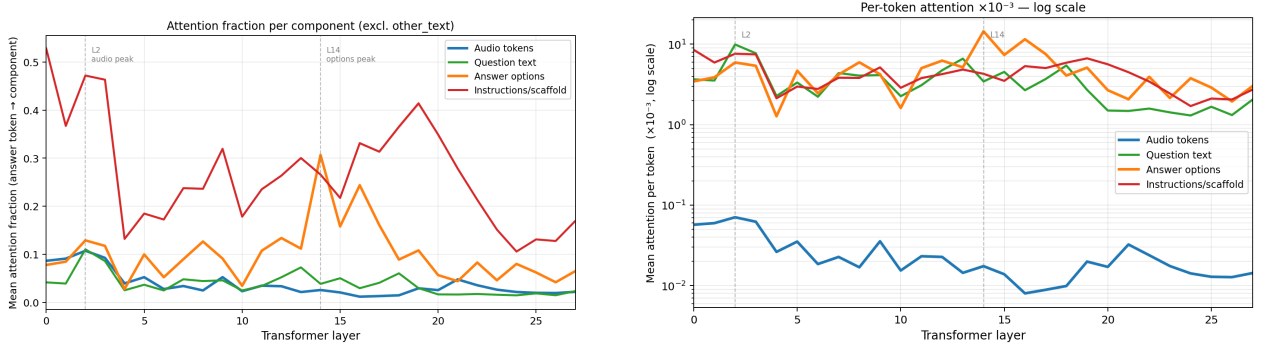
The section includes three complementary experiments. The first analyses attention patterns, measuring how the response token distributes its attentional budget across audio, question, options, and prompt instructions. The second applies a *logit lens* to the representations of audio tokens, in order to verify whether they preserve lexically interpretable content across the model layers. The third uses an *attention knockout* procedure, selectively blocking specific attentional flows to estimate the causal role of the different input components in response generation.

All experiments are conducted on the *Thinker* component, a 28-layer multimodal transformer. HumMusQA is evaluated on 960 instances, obtained from the 320 original samples through three permutations of the answer-option order. This choice reduces the risk of biases related to the position of the correct answer, which in the original version of the dataset always occupies position A. The effect of the permutations is nevertheless limited: 81.9% of the samples produce the same response independently of the option order, and accuracy remains almost unchanged, moving from 66.56% in the original configuration to 66.46% in the permuted configuration. This confirms that the subsequent analyses are not dominated by instabilities due to the ordering of the alternatives. One final element is crucial for interpreting the results: audio tokens represent, on average, 93.1% of the input sequence, whereas all textual tokens constitute only the remaining 6.9%. Consequently, any low attentional or causal weight assigned to audio cannot be attributed to its absence from the context, but must be interpreted with respect to a strong structural asymmetry between the amount of available audio tokens and their actual use by the model.

4.4.1 Attention Pattern

The experiment measures, for each transformer layer (L0–L27), the fraction of attention that the generated response token assigns to the main groups of input tokens: *audio*, *instruction*, *question*, *options*, and *other_text*. The measurement is performed at the moment of generation of the first response token, corresponding to the selected letter in the multiple-choice task (A/B/C/D). Distinguishing textual tokens according to their role in the prompt makes it possible to separate the contribution of the question, the options, and the instructional scaffold.

The main result is the structural marginality of audio, already visible in Fig. 12. Although audio tokens constitute approximately 93% of the entire input sequence, the generated token assigns them only a minimal share of the attentional budget. The mean attention towards audio is equal to 3.8% of the total and, once normalised per token, is approximately two orders of magnitude lower than that assigned to textual tokens.



(a) Fraction of attentional budget per component (excl. **other_text**) across the 28 transformer layers. Audio peaks at L2 (10.7%) then stabilises below 3%; options show a transient peak at L14 (30.7%).

(b) Mean attention per token on a logarithmic scale. Audio (blue) sits 1–2 orders of magnitude below all textual components at every layer.

Figure 12: Layer-wise attention of the generated answer token to each input component ($N = 960$ evaluations, 28 layers). Audio tokens, despite comprising $\approx 93\%$ of the input sequence, receive only 3.8% of the total attentional budget; per-token audio attention is $\approx 160\times$ lower than instruction tokens already at L0.

Subfigure 12a shows the aggregate attentional budget per component, whereas Subfigure 12b reports the mean attention per token. This second normalisation is particularly important: although audio occupies approximately 1590 tokens on average, the share of attention it receives in the first layer is only 8.6%, corresponding to a mean attention per token of approximately 5.4×10^{-5} . Already in the initial layer, each token belonging to the *instruction* component receives approximately 160 times more attention than a single audio token.

The layer-wise trend confirms this asymmetry. Audio reaches its maximum at L2, with 10.7% of the attentional budget, but then stabilises around a mean of 2.81% for the rest of the forward pass. The *instruction* component instead receives an initially very high share, equal to 52.82%, which progressively decreases until L24 (10.56%) and partially rises again in the final layers, reaching 16.93% at L27. The *options* show a different behaviour: they receive little attention in the initial and final phases, but exhibit a marked peak between L14 and L16, reaching 30.67% at L14 and 24.37% at L16, before returning below 10% from L19 onward. This peak, the maximum observed for a variable informational component of the prompt, suggests a central computational window devoted to the comparative evaluation of the answer alternatives.

The **other_text** component, associated with invariant scaffold and prompt-format tokens, instead follows an increasing trajectory from L4 onward and reaches its maximum at L26, with 79.65% of the attentional budget. A clear dynamic therefore emerges: the central layers show a transient phase of option reading, whereas the late layers are increasingly dominated by fixed text and prompt format.

Aggregating the layers into three macro-regions, *Early* (L0–L9), *Mid* (L10–L18), and *Late* (L19–L27), this trajectory becomes even more evident (Table 1).

Table 1: Mean attention received by each input component across the three depth regions of the model.

Component	Early	Mid	Late	Change E→L
Audio	0.0607	0.0220	0.0274	−3.3 pp
Instruction	0.3112	0.2744	0.2155	−9.6 pp
Question	0.0501	0.0447	0.0185	−3.2 pp
Options	0.0895	0.1494	0.0651	−2.4 pp (+Mid)
Other text	0.4884	0.5094	0.6735	+18.5 pp
All text	0.9393	0.9780	0.9726	+3.3 pp

The most relevant aspect of Table 1 is not only the predominance of text, but the internal redistribution of its components. The task-related textual components of the prompt, namely *instruction* (−9.6 pp), *question* (−3.2 pp), and *options* (−2.4 pp, despite the peak in the *mid* phase), lose weight towards the deeper layers, whereas **other_text** gains 18.5 percentage points. As the network progresses, the model therefore appears to shift from processing variable content towards the fixed format of the prompt. Audio, already marginal in the *early* region with 6.1% of the attentional budget, drops to 2.7% in the *late* region, distributed over approximately 1590

tokens and therefore associated with substantially negligible per-token attention.

The comparison between the 638 correct and 322 incorrect evaluations reveals two secondary but coherent signals. For the audio component, the *correct-wrong* delta is null in the *early* region ($-0.0003 \approx 0$), but becomes positive from L8 onward, reaching its maxima at L14 ($+0.0038$) and L19 ($+0.0036$). Correct responses therefore show a slight but systematic prevalence of attention towards audio in the *mid* and *late* regions, with an effect of approximately 0.2–0.4 percentage points. For the *options*, instead, the signal has the opposite direction: incorrect responses assign greater attention to the options during the *mid* phase ($\Delta = -0.0069$), with the largest gap at L14 (-0.0176), namely in the layer of the absolute *options* peak; high values are also observed at L12 (-0.0082), L13 (-0.0085), and L16 (-0.0099). These results suggest a possible mechanism of *answer picking by surface form*: when the model is wrong, it tends to focus more strongly on the textual form of the answer alternatives, rather than on a more balanced integration of the context.

The analyses by difficulty and category confirm that the attentional pattern is substantially stable. Despite an accuracy gap of approximately 19 percentage points between the *Low* and *High* classes, the attention curves are almost overlapping across all 28 layers: the model does not increase attention towards audio when the question becomes more difficult, nor does it noticeably modify attention towards options or instructions. Similarly, no clear correlation emerges between category accuracy and attention to audio. The *Sound Texture* category, for example, presents the lowest *audio_late* value in the entire dataset (2.15%) and the highest text/audio ratio ($50.9\times$), while reaching an accuracy of 76.5%, above the average. The only partial exception is *Lyrics*, which shows an *audio_early* value higher than the average (7.69% compared with approximately 6.1%), likely due to the greater recognisability of sung linguistic content; however, this difference disappears in the *late* region, where audio attention returns to 2.73%.

Overall, the experiment shows that the decision token almost always operates in an attention regime dominated by text. Audio is massively present in the input, but receives marginal attentional weight and does not increase either with difficulty or in the categories where stronger acoustic grounding would be expected. The central peak on the *options* instead suggests that a relevant part of the decision-making process occurs as a comparison between textual alternatives, consistently with the text-first behaviour observed in the post-hoc experiments.

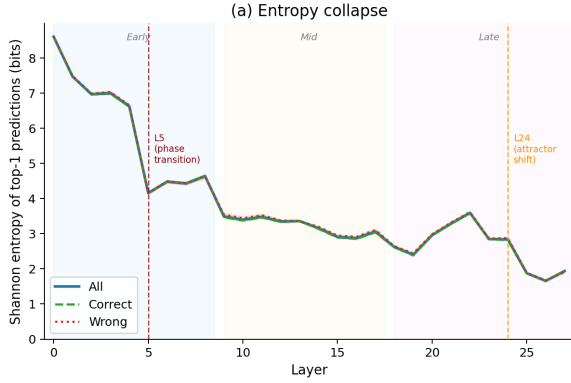
4.4.2 Logit Lens

The *Logit Lens* makes it possible to project the *hidden states* of audio tokens, at each transformer layer, through the model’s linguistic head (*LM head*). The latter is the same linear layer used by the model to generate the next token: it transforms internal representations that are not directly interpretable into a probability distribution over the entire vocabulary. Applying this projection to intermediate layers therefore makes it possible to observe which tokens the audio representations would correspond to if processing were interrupted at a specific point in the network.

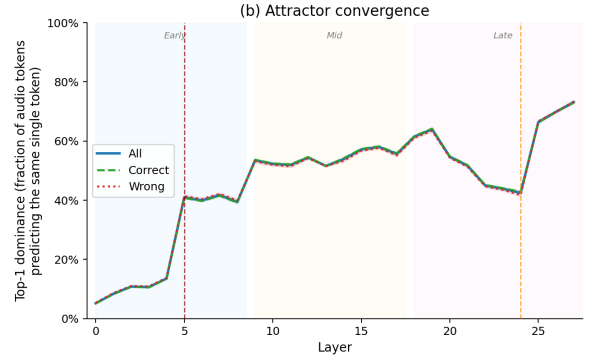
For each projection, two complementary metrics are computed. **Shannon entropy** H measures the dispersion of the distribution over the vocabulary: high values indicate representations that are still diffuse, whereas low values indicate concentration on a few tokens. **Top-1 Dominance** instead measures the fraction of audio tokens for which the same vocabulary token is the most likely prediction. High values indicate the presence of an *attractor*, namely an internal dynamic that causes many audio representations to converge towards the same lexical prediction, independently of the original acoustic content.

This analysis is purely observational: it does not modify the model’s behaviour and does not directly measure the causal role of audio representations in the final decision. Instead, it describes how these representations evolve internally during the forward pass. Considering 960 evaluations, an average of 1589.6 audio tokens per sample, and 28 layers of the *Thinker*, the experiment involves approximately 42.7 million projections.

The main result is a progressive collapse of audio representations towards a few lexical *attractors*. As shown in Fig. 13, Subfigure 13a shows that at layer L0 the audio representations are still diffuse, with entropy $H = 8.61$ bits, equivalent to approximately 383 effective tokens. At layer L26, entropy drops to $H = 1.66$ bits, corresponding to fewer than 4 effective tokens. In parallel, Subfigure 13b shows that *Top-1 Dominance* increases from 5.2% to 69.9%, indicating that almost 70% of audio tokens are projected onto the same vocabulary token. The reduction of nearly 7 bits in entropy represents the quantitatively strongest result of the entire analysis.



(a) Shannon entropy (bits) of the projected vocabulary distribution across the 28 layers. Entropy drops from 8.61 bits at L0 to 1.66 bits at L26, with the sharpest phase transition at L4–5; correct and wrong responses follow nearly identical trajectories.



(b) Top-1 Dominance (fraction of audio tokens projecting to the same most likely vocabulary token) across layers. Dominance rises from 5.2% at L0 to 69.9% at L26; curves for correct and wrong responses are essentially overlapping.

Figure 13: Logit Lens analysis of audio-token representations across the 28 *Thinker* layers ($N = 960$ evaluations, ≈ 42.7 M projections). Shannon entropy collapses from 8.61 to 1.66 bits between L0 and L26, with a sharp phase transition at L4–5; Top-1 Dominance rises from 5.2% to 69.9%, indicating that by the late layers nearly 70% of all audio tokens project onto the same vocabulary token.

The collapse is not uniform, but occurs through two main transitions and an unexpected dynamic in the final phase. The first transition is observed between L4 and L5 and is the most pronounced change of the entire forward pass: the number of effective tokens decreases from approximately 100 to approximately 17, entropy is nearly halved, and *dominance* triples. After L5, the audio representations no longer return to levels comparable with the initial ones, suggesting an architectural point of no return at which the internal reference system associated with audio is stabilised. The abruptness of the transition suggests a structural compression or routing function, rather than a gradual transformation.

The second transition occurs after a short plateau between L5 and L8 and produces a further contraction of the representational space. In this region, *Top-1 Dominance* exceeds 50% for the first time, indicating that more than half of the audio tokens now share the same dominant prediction. From that point onward, the residual information appears strongly concentrated.

In the *late* phase, instead, a non-monotonic behaviour emerges. Between L19 and L22, entropy temporarily increases before decreasing again in the final layers, forming a double-valley structure. This rebound suggests a possible temporary redistribution of the audio representations: one possible interpretation is the reinjection of information from text through attention mechanisms or MLP blocks; an alternative hypothesis is competition between different *attractors*, resolved only in the final layers. In the absence of dedicated causal analyses, this phenomenon remains an open empirical observation. The complete layer-wise values, distinguishing between correct and incorrect responses, are reported in Table A.2 in Appendix A; a summary by depth regions is reported in Table 2.

Table 2: Mean entropy and dominance across the three depth regions of the model.

Phase	Range	Mean H	Mean dominance	ΔH vs Early
Early	L0–L8	6.045	23.38%	—
Mid	L9–L17	3.232	54.26%	−2.81 bits
Late	L18–L27	2.603	57.26%	−3.44 bits

Once the collapse has been characterised, it is necessary to observe which tokens the audio representations converge towards. The *attractor* system presents a dual structure. The token 讓他們 (Chinese: “let them”) dominates in the *early* and *mid* regions, where it collects almost 60% of the projections, before decreasing to approximately 40% in the *late* phase. The token ă, a Vietnamese sub-syllable, is instead the second stable *attractor*: almost irrelevant in the *early* and *mid* regions, with a *dominance* close to 8%, it progressively grows in the final phase until reaching 34%.

Overall, in the *late* phase these two tokens alone absorb 74.1% of more than 14 million projections. The remaining 26% is distributed across dozens of different tokens, consisting of lexical fragments belonging to

languages such as Arabic, Japanese, Polish, and Traditional Chinese, without any evident relation to the musical content, the dataset categories, or the content of the questions. The aggregate values are reported in Table 3.

Table 3: Concentration of the main attractors across the three depth regions of the model.

Metric	Early	Mid	Late
Top-1 (讓他們)	36.2%	59.2%	40.1%
Top-2 (Á)	8.6%	8.3%	34.0%
Cumulative Top-2	44.8%	67.5%	74.1%

The absence of semantic relevance in the *attractors* is the most significant observation of the experiment. The internal vocabulary towards which the audio representations converge shows no interpretable connections with the acoustic signal, the musical track, or the associated question. One possible explanation derives from the architecture of Qwen2.5-Omni-7B: audio is first transformed into a mel-spectrogram, then processed by Whisper-large-v3, a 32-layer transformer encoder, and only subsequently projected into the embedding space of the *Thinker*. It is therefore plausible that, when the audio tokens reach L0, a substantial part of the acoustic information has already been compressed into abstract representations that cannot be recovered through a simple lexical projection with the *Logit Lens*.

The control analyses on correctness, category, and difficulty all converge towards the same conclusion: the collapse towards non-semantic *attractors* is a systematic phenomenon. The comparison between the 638 correct and 322 incorrect evaluations shows almost overlapping entropy curves across all 28 layers and practically identical final values ($H_{\text{correct}} = 2.594$, $H_{\text{wrong}} = 2.622$). The thirteen musical categories also start from high initial values ($H \approx 8.6$ bits) and converge towards the same final interval ($H \approx 1.55\text{--}1.77$ bits), without distinctive profiles. Even categories that should theoretically require greater use of audio information, such as *Sound Texture* and *Functional Context*, follow the same trajectory. The only slight exception is *Lyrics*, which maintains slightly higher entropy in the intermediate layers, probably due to the presence of linguistic content in sung audio, but this difference disappears in the final phase. The same holds for difficulty: the curves for the *Low*, *Medium*, and *High* levels are almost perfectly overlapping and differ at L26 by only 0.005 bits overall.

Overall, the *Logit Lens* shows that audio representations in the *Thinker* rapidly collapse towards a few non-semantic lexical *attractors*. This collapse does not depend on response correctness, musical category, or question difficulty, but appears as a stable behaviour deeply rooted in the model’s internal architecture. The result does not necessarily imply that audio information is absent, but indicates that, once it enters the *Thinker*, it does not remain accessible in a lexically interpretable form through the projection of the linguistic head.

4.4.3 Attention Knockout

The experiment uses the logic of *attention knockout* to estimate the causal dependence of the response on the different groups of input tokens. Operationally, a *forward hook* is applied to the *self-attention* module of each transformer layer, selectively nullifying the attention of the decision token towards a specific group of tokens. The decision token corresponds to the last token of the prompt, namely the token from which the first letter of the response is generated. Within the considered layer window, this token therefore cannot receive information from the masked group, since the corresponding attention scores are masked before the softmax, preventing the selected tokens from contributing to the attention output.

The block is applied during the *prefill*, when the model processes the entire input sequence before generation; subsequently, generation proceeds normally. For each component–window pair, the response produced under baseline conditions is compared with the one obtained after the knockout. The **flip rate**, namely the frequency with which the response changes, the direction of the change, harmful (*good*→*bad*) or beneficial (*bad*→*good*), and the net variation in accuracy, denoted by Δacc , are then measured.

The five analysed components cover the main parts of the input: **audio**, namely the audio tokens produced by Whisper, approximately 1590 per sample and corresponding to about 93% of the input; **question**, namely the tokens of the musical question, approximately 11; **options**, namely the tokens of the A/B/C/D answer options, approximately 23; **instruction**, namely the tokens of the task scaffold, approximately 63; and **all_text_to_audio**, which simultaneously blocks the attention of all textual tokens towards audio, with the objective of isolating audio from the surrounding textual context. The knockout is applied over 14 windows of 2 layers each, covering all 28 layers of the *Thinker*.

The main result is consistent with the previous experiments: blocking the decision→audio channel changes the response in only **0.737%** of cases, with an accuracy variation equal to $\Delta acc = -0.141\%$. Audio is therefore the least influential component among those analysed.

Table 4: Summary of the attention knockout results for each input component.

Component	Flip rate	Ko accuracy	Δacc	Ko corr base corr	Peak window	Peak flip	Peak Δacc
audio	0.737%	66.317%	-0.141%	99.552%	L6-L8	1.042%	-0.417%
question	0.737%	66.347%	-0.112%	99.563%	L12-L14	0.729%	-0.417%
options	1.734%	66.280%	-0.179%	99.015%	L18-L20	3.542%	-0.625%
instruction	1.905%	66.004%	-0.454%	98.657%	L24-L26	6.354%	-1.458%
all_text_to_audio	3.631%	65.729%	-0.729%	97.582%	L14-L16	16.354%	-5.833%

Table 4 shows that blocking the *instruction* produces an effect approximately three times larger than directly blocking audio ($\Delta acc = -0.454\%$), whereas the simultaneous *all_text_to_audio* block causes an accuracy reduction of -0.729% , approximately five times larger than the effect of audio, while still remaining limited in absolute terms. The direct audio channel is therefore the least causally relevant, even compared with the question and the instructions. If the model were directly exploiting the audio signal to answer, one would expect a flip rate at least comparable to that of the textual components, but this does not occur.

Two readings of the table are particularly informative. The **Ko corr | base corr** column indicates how often the knockout preserves an already correct response: for audio, the value is 99.552%, so almost all correct responses remain correct even when the decision→audio channel is blocked. The **Ko corr | base wrong** column instead measures how often the knockout corrects initially wrong responses: for audio, the value is only 0.47%. Audio knockout therefore produces an almost null effect, both positively and negatively, suggesting that the direct audio channel is functionally almost inert with respect to the final decision.

Table 5 shows how this effect is distributed along the depth of the network.

Table 5: Flip rate of attention knockout by layer window and input component.

Window	audio	question	options	instruction	all_text_to_audio
L0-2	0.52%	0.83%	0.63%	0.73%	4.38%
L2-4	0.83%	0.63%	0.73%	0.83%	2.08%
L4-6	1.04%	0.52%	0.83%	0.63%	3.23%
L6-8	1.04%	0.94%	0.73%	1.04%	2.60%
L8-10	0.63%	0.63%	0.52%	0.83%	6.04%
L10-12	0.83%	0.83%	0.63%	0.52%	3.02%
L12-14	0.83%	0.73%	0.63%	1.35%	4.79%
L14-16	0.73%	0.63%	0.63%	2.40%	16.35%
L16-18	0.42%	0.73%	1.35%	3.33%	3.65%
L18-20	0.73%	1.67%	3.54%	2.08%	1.35%
L20-22	0.73%	0.42%	3.33%	2.08%	1.15%
L22-24	0.63%	0.73%	2.92%	1.88%	0.83%
L24-26	0.63%	0.73%	5.42%	6.35%	0.52%
L26-28	0.73%	0.31%	2.40%	2.60%	0.83%

Distinct profiles emerge from the layer-wise distribution. Audio is the only component with a substantially flat trend: values oscillate between 0.42% and 1.04%, without a clear peak or privileged window. The only slight increase occurs between L4 and L8, but the effect is too small to indicate a true phase of audio integration. The *question* shows a similar profile, with a single anomaly at L18-L20 (1.67%), probably noisy given the small number of tokens involved, approximately 11. The linguistic question therefore does not appear to have a specific integration window.

The *options*, by contrast, present a more defined pattern: the effect is negligible until L16 and increases in the *late* phase, reaching a peak at L24-L26 of 5.42%, with $\Delta acc = -0.625\%$. This suggests that the model refines the discrimination between A/B/C/D in the final phases of computation. The *instruction*, instead, shows a double-peak structure: it remains almost inactive until L12, with values around 0.5-1%, increases at L16-L18 (3.33%), and reaches its maximum at L24-L26 (6.35%). This suggests a dual function: in the intermediate layers, the instruction contributes to the integration of the MCQ format, whereas in the final layers it reinforces the output constraint, namely the production of a single letter.

The *all_text_to_audio* component is the most informative: it already shows high values in the initial phase, reaches 6.04% at L8-L10, increases sharply at L14-L16 with a flip rate of 16.35%, and rapidly decays after L18. The peak at L14-L16 causes an accuracy drop of almost 6 percentage points; moreover, 55% of flips occur in the harmful direction, from correct to wrong, whereas only 20% occur in the opposite direction. This rules out

a purely random explanation and indicates that blocking the text→audio channel is causally harmful in the central phase of the network.

Interpreting the peak requires linking the knockout to the Attention Pattern. The L14–L16 window indeed coincides with the point at which the attentional analysis records the maximum attention to the *options*, equal to 30.7% at L14. In these layers, the decision token appears to actively discriminate between the answer alternatives, constructing a representation that indirectly uses the audio tokens as a contextual background. When the entire text→audio channel is blocked, and not only the decision→audio channel, this global contextual integration is interrupted and the model loses computational coherence at the critical moment of selection.

This interpretation is reinforced by the asymmetry with respect to direct audio blocking: in the same interval, blocking only the decision→audio channel produces marginal effects, around 0.7% of flips. It therefore does not seem to be the audio signal itself that determines the decision, but rather the overall structure of the attentional field, which degrades when all textual pathways towards audio are removed simultaneously.

The aggregation by phase confirms this functional division (Table 6). Audio is the only component with a Δacc that decreases monotonically from the *early* phase to the *late* phase: already weakly influential in the initial layers, it becomes progressively more inert along the depth of the network. The residual effect observed in *early* may reflect the initial encoding of audio features, but is too small to be interpreted as semantic integration. By contrast, *all_text_to_audio* concentrates its effect in the *mid* phase, with $\Delta acc = -2.109\%$, and decays almost completely in the *late* phase, where it reaches only -0.062% . The *options* and *instruction* components instead follow the opposite trajectory, increasing towards the *late* phase: the intermediate layers therefore appear to integrate the overall context, including the multimodal component, whereas the final layers consolidate the discrimination between options and the MCQ response format.

Table 6: Flip rate and accuracy variation of attention knockout aggregated across the three depth regions of the model.

Component	Early flip	Early Δacc	Mid flip	Mid Δacc	Late flip	Late Δacc
audio	0.813%	−0.292%	0.703%	−0.078%	0.688%	−0.042%
question	0.708%	−0.104%	0.729%	−0.182%	0.771%	−0.062%
options	0.688%	−0.083%	0.807%	−0.130%	3.521%	−0.312%
instruction	0.813%	−0.250%	1.901%	−0.495%	3.000%	−0.625%
all_text_to_audio	3.667%	−0.292%	6.953%	−2.109%	0.937%	−0.062%

For all components, the direction of the effect is predominantly negative: knockout produces more harm than benefit. For audio, 103 flips are observed across all layer windows, corresponding to a very small fraction of the component-window evaluations. At the sample level, only 20 out of 320 samples, equal to 6.2%, perform at least one flip in any window when audio is blocked. For *all_text_to_audio*, instead, the number rises to 135 samples, equal to 42.2%, confirming that a substantial share of examples is sensitive to the global interruption of the multimodal channel, even though the direct audio channel is almost always uninfluential.

A further result concerns the *instruction* component. Its *net harm* is equal to +52, second only to that of *all_text_to_audio* (+86), despite presenting fewer total flips than the *options*. When the instruction is blocked, the probability that the flip is harmful is therefore higher: the instruction plays a more critical role for response correctness than the options. Moreover, the peak at L24–L26, with a flip rate of 6.35% and $\Delta acc = -1.458\%$, represents the second most harmful window of the entire experiment. The instruction therefore does not operate as a monolithic signal, but appears to activate at two distinct moments: an intermediate phase, associated with the integration of the MCQ format, and a late phase, linked to the consolidation of the output constraint. The fact that output parsing almost always remains valid suggests that, even when this channel is blocked, the model preserves the ability to produce an answer in the correct format, but loses part of the contextual information needed to select the correct response.

Overall, attention knockout confirms the interpretation that emerged from the previous experiments, but with more direct causal evidence. The decision→audio channel is almost inert, whereas the global interruption of the text→audio channel in the intermediate layers is substantially more harmful. The model therefore does not seem to base the final choice on direct access to audio tokens, but on a more indirect multimodal configuration, in which audio acts as a contextual background during the central phase of comparison between textual alternatives. The final decision, instead, is consolidated mainly through the options and the instruction in the deeper layers.

5. Conclusions

This thesis addressed a central question in the evaluation of Large Audio Language Models: when a model answers a question about music, does it ground its response in the acoustic signal, or does it mainly exploit linguistic and musical knowledge acquired during pretraining? The experiments on Qwen2.5-Omni-7B show that accuracy alone cannot answer this question. A correct response is not necessarily evidence of listening: it is necessary to distinguish between the presence of audio in the input, its internal representation, its accessibility to the language model, and its causal role in the final decision.

The results show that audio is structurally processed, but only weakly used as an independent decision source. The Logit Lens analysis reveals that audio-token representations undergo a strong internal reorganisation inside the *Thinker*: their entropy rapidly collapses and a small number of non-semantic lexical *attractors* dominates the late layers, with the two main attractors accounting for 74.1% of the dominant late-stage projections. This does not imply that the audio representation is empty. Since the waveform is first processed by Whisper-large-v3 and only then projected into the *Thinker*, relevant acoustic information may already be encoded in a compact and abstract form, not recoverable through a direct projection onto the linguistic vocabulary. Logit Lens therefore does not prove the absence of musical information, but shows that, once audio enters the *Thinker*, it is no longer accessible in a lexically interpretable form.

The attention-based analyses clarify how this representation is used. Although audio tokens constitute approximately 93% of the input sequence, the generated answer token assigns them only 3.8% of the total attentional budget, and the per-token attention to audio remains one to two orders of magnitude below that assigned to textual components. This could be compatible with the hypothesis that Whisper has already compressed the relevant acoustic information and that the *Thinker* only needs limited access to it. However, the attention knockout experiment weakens this interpretation: blocking the direct decision→audio channel changes the answer in only 0.737% of cases and reduces accuracy by only −0.141%. Thus, whatever information is encoded by the audio encoder, the final decision rarely depends on direct access to audio tokens.

The most informative exception occurs in the intermediate layers, especially around L14–L16. This is the same region in which attention to the answer options reaches its maximum, suggesting a phase of active comparison between multiple-choice alternatives. In this window, blocking the entire `all_text_to_audio` pathway produces a much stronger effect, with a flip rate of 16.35% and an accuracy drop of −5.833%, whereas blocking only the direct decision→audio channel remains almost irrelevant. This asymmetry indicates that audio does not mainly function as an autonomous perceptual source retrieved at decision time. Instead, it contributes indirectly to the global attentional context in which the question and answer options are organised.

The attribution-based analyses independently support the same interpretation. MM-SHAP shows a predominantly text-first profile: as task difficulty increases, the relative contribution of audio decreases, while the contribution of text increases. audioDIME further separates unimodal contributions from multimodal interactions: the unimodal text component remains dominant, whereas the audio contribution is weak when considered in isolation. At the same time, correct responses are associated with stronger multimodal interaction. Audio is therefore not completely ignored, but its role is mostly mediated by the textual question and by the structure of the multiple-choice alternatives.

Taken together, the experiments converge on a coherent conclusion. Qwen2.5-Omni-7B does process audio, and its encoder plausibly produces rich acoustic representations. However, on HumMusQA, the *Thinker* rarely uses those representations as a direct causal basis for the final answer. The model can often rely on textual priors, including knowledge about instruments, genres, harmonic conventions, musical forms, and cultural associations. This explains why removing or blocking direct access to audio has limited effects, while perturbing textual structure or disrupting multimodal interaction has stronger consequences.

The main contribution of this thesis is therefore not simply to observe a textual bias, but to disentangle different meanings of “using audio”. Audio can be encoded without being lexically interpretable, present without being causally decisive, and involved in a multimodal interaction without acting as an independent perceptual ground. This distinction is essential for evaluating audio-language models. As long as MQA benchmarks remain partially solvable through linguistic priors, accuracy will tend to overestimate genuine listening. Reliable evaluation should therefore combine performance metrics with causal interventions and disentangled interpretability analyses, measuring not only whether a model answers correctly, but how that answer is produced.

6. Methodological Limitations

The proposed methodology has several intrinsic limitations. First, MM-SHAP and audioDIME are computationally expensive, since they require many forward passes over perturbed inputs and their cost grows with the number of interpretable features. Caching, multi-GPU parallelisation, perturbation reuse, and efficient DIME matrix updates reduce this burden, but do not remove the constraint imposed by available resources.

Second, the explanations depend on how audio is made interpretable. The audioLIME-inspired representation, based on source-segment features, is more musically meaningful than uniform temporal windows, but remains sensitive to source separation and segmentation choices. Long segments may mix musical events, short segments increase instability and cost, and the use of eight onset-based segments is therefore a compromise between readability, local faithfulness, and feasibility. Similarly, Demucs makes perturbations more interpretable, but introduces dependence on stem quality: leakage, artefacts, incomplete separation, or different choices in the number of stems would change the feature space and, consequently, the explanations.

A further limitation concerns masking. Removing, attenuating, or replacing a feature is only an approximation of the absence of the corresponding information. In audio, perturbations may introduce artefacts or unnatural distributions; in text, word-level masking improves readability but does not fully remove subword tokenisation effects or the possibility of locally unnatural prompts. In audioDIME, the estimation of UC and MI also depends on the background distribution used to approximate marginal expectations, so these quantities should be interpreted as empirical estimates conditioned on the chosen reference samples, not as absolute model properties. Finally, the results are specific to Qwen2.5-Omni-7B, to the *Thinker* component, and to the HumMusQA multiple-choice setting. They cannot be automatically generalised to other Audio LLMs, different encoders, fusion strategies, or open-ended music captioning. Moreover, MM-SHAP, LIME, and audioDIME remain local post-hoc methods: they describe model behaviour under controlled perturbations, but do not provide direct access to internal representations or prove musical understanding in a human sense. The results should therefore be read as a controlled analysis of multimodal behaviour in Music Question Answering, rather than as a complete account of audio-language understanding.

7. Future Work

Future work may first extend the analysis from controlled Music Question Answering to open-ended music captioning. In that setting, the model freely generates a description rather than selecting among A/B/C/D options, so the target can no longer be defined only through option logits. The methodology would need scoring functions for generative sequences, such as semantic similarity, multiple reference captions, or concept-specific musical scores, together with datasets designed to reduce textual shortcuts.

A second direction is model comparison. The behaviour observed here may depend on the architecture of Qwen2.5-Omni-7B, including its audio encoder, alignment mechanism, and *Thinker–Talker* structure. Applying the same pipeline to Audio LLMs and more general Multimodal LLMs with different encoders, fusion strategies, cross-attention mechanisms, pretraining data, or musical specialisation would clarify whether the weak causal role of audio is a general tendency or an architectural consequence.

A third direction concerns datasets. HumMusQA provides a rigorous musical benchmark, but remains multiple-choice and can still contain textual shortcuts. Testing MM-SHAP, audioDIME, and the internal analyses on captioning datasets, alternative MQA benchmarks, counterfactual datasets, audio-only QA datasets, or collections balanced by genre, culture, and difficulty would help distinguish genuine audio grounding from linguistic priors and musical or cultural biases.

Finally, future work should refine the interpretable representation of audio itself, testing different temporal granularities, numbers of stems, source-separation methods, and perceptually controlled masking strategies. This would make it possible to evaluate how much explanations depend on the chosen feature space and to make audioDIME more robust, scalable, and suitable for complex generative scenarios.

In conclusion, the question posed by Turing remains relevant, but must now be extended beyond imitation. It is not enough to observe that a machine produces a correct or plausible answer: we must ask how that answer is generated. For Large Audio Language Models, this means distinguishing genuine grounding in the acoustic signal from shortcuts, statistical memories, and knowledge acquired during training. Explainable AI does not solve neural opacity, but it makes model behaviour more readable. In this sense, interpreting a model also means returning to the deeper question of what it means to understand, perceive, and reason, both in machines and, by reflection, in human beings.

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A. Appendix A

Table 7: Mean attention received by each input component at each model layer.

L	Audio	Instruction	Question	Options	Other text	All text
0	0.0864	0.5282	0.0415	0.0777	0.2662	0.9136
1	0.0909	0.3671	0.0390	0.0846	0.4184	0.9091
2	0.1073	0.4718	0.1103	0.1289	0.1817	0.8927
3	0.0925	0.4632	0.0859	0.1174	0.2410	0.9075
4	0.0393	0.1320	0.0250	0.0277	0.7760	0.9607
5	0.0523	0.1847	0.0366	0.0996	0.6268	0.9477
6	0.0276	0.1722	0.0247	0.0522	0.7233	0.9724
7	0.0339	0.2376	0.0480	0.0897	0.5909	0.9661
8	0.0247	0.2362	0.0442	0.1265	0.5684	0.9753
9	0.0523	0.3194	0.0457	0.0910	0.4916	0.9477
10	0.0231	0.1783	0.0247	0.0343	0.7396	0.9769
11	0.0346	0.2352	0.0340	0.1073	0.5889	0.9654
12	0.0335	0.2636	0.0521	0.1338	0.5171	0.9665
13	0.0214	0.3002	0.0727	0.1117	0.4940	0.9786
14	0.0256	0.2658	0.0384	0.3067	0.3636	0.9744
15	0.0205	0.2172	0.0501	0.1579	0.5542	0.9795
16	0.0118	0.3312	0.0294	0.2437	0.3840	0.9883
17	0.0130	0.3134	0.0406	0.1606	0.4724	0.9870
18	0.0146	0.3653	0.0603	0.0889	0.4709	0.9854
19	0.0291	0.4140	0.0295	0.1080	0.4194	0.9709
20	0.0254	0.3491	0.0165	0.0566	0.5524	0.9746
21	0.0477	0.2783	0.0163	0.0439	0.6138	0.9523
22	0.0355	0.2132	0.0173	0.0828	0.6512	0.9645
23	0.0263	0.1514	0.0158	0.0459	0.7606	0.9737
24	0.0215	0.1056	0.0145	0.0799	0.7784	0.9785
25	0.0197	0.1309	0.0185	0.0621	0.7687	0.9803
26	0.0194	0.1276	0.0147	0.0418	0.7965	0.9806
27	0.0217	0.1693	0.0234	0.0649	0.7207	0.9783

Table 8: Aggregate results by musical category. For each category, the number of samples, accuracy, audio attention in the *early* and *late* regions, attention to the options in the *mid* region, and the ratio between textual and audio attention in the *late* phase are reported.

Categoria	N	Accuracy	Audio Early	Audio Late	Options Mid	Text/Audio Late
Functional Context	48	91.7%	0.0601	0.0248	0.1443	46.3×
Mood and Expression	63	79.4%	0.0625	0.0219	0.1509	48.2×
Sound Texture	81	76.5%	0.0564	0.0215	0.1504	50.9×
Genre and Style	66	69.7%	0.0651	0.0251	0.1465	44.3×
Historical & Cultural	39	69.2%	0.0642	0.0278	0.1491	39.0×
Instrumentation	189	66.7%	0.0630	0.0297	0.1403	37.5×
Musical Texture	33	66.7%	0.0591	0.0250	0.1505	45.5×
Lyrics	33	66.7%	0.0769	0.0273	0.1622	37.8×
Performance	75	64.0%	0.0621	0.0281	0.1490	42.8×
Structure	21	61.9%	0.0553	0.0231	0.1533	44.0×
Melody	102	60.8%	0.0590	0.0253	0.1543	43.7×
Harmony	108	60.2%	0.0579	0.0263	0.1573	40.7×
Metre and Rhythm	102	50.0%	0.0638	0.0263	0.1533	43.3×

Table 9: Entropy and dominance metrics across transformer layers. H denotes the entropy of the logit distribution, while Dominance represents the probability mass assigned to the most probable token. Correct and wrong refer to samples answered correctly or incorrectly by the model.

L	H_{all}	H_{correct}	H_{wrong}	ΔH (c-w)	Dominance	Dom _c	Dom _w	ΔDom
0	8.610	8.607	8.617	-0.010	5.16%	5.08%	5.31%	-0.23 pp
1	7.476	7.470	7.489	-0.019	8.36%	8.22%	8.62%	-0.40 pp
2	6.970	6.960	6.991	-0.031	10.79%	10.70%	10.98%	-0.29 pp
3	7.004	6.988	7.034	-0.045	10.62%	10.49%	10.88%	-0.39 pp
4	6.628	6.614	6.654	-0.040	13.52%	13.41%	13.73%	-0.31 pp
5	4.160	4.163	4.155	+0.008	40.95%	40.74%	41.36%	-0.61 pp
6	4.484	4.485	4.483	+0.001	39.93%	39.71%	40.37%	-0.67 pp
7	4.433	4.432	4.434	-0.002	41.71%	41.50%	42.12%	-0.62 pp
8	4.643	4.651	4.628	+0.023	39.41%	39.16%	39.90%	-0.74 pp
9	3.493	3.468	3.540	-0.072	53.50%	53.67%	53.15%	+0.52 pp
10	3.399	3.377	3.441	-0.063	52.27%	52.43%	51.97%	+0.46 pp
11	3.484	3.463	3.525	-0.062	51.90%	52.14%	51.42%	+0.72 pp
12	3.350	3.337	3.376	-0.038	54.45%	54.57%	54.20%	+0.37 pp
13	3.361	3.362	3.358	+0.005	51.58%	51.60%	51.55%	+0.04 pp
14	3.157	3.138	3.194	-0.056	53.88%	54.16%	53.33%	+0.82 pp
15	2.910	2.892	2.946	-0.054	57.13%	57.34%	56.71%	+0.63 pp
16	2.874	2.853	2.914	-0.060	58.02%	58.20%	57.65%	+0.55 pp
17	3.062	3.040	3.106	-0.066	55.63%	55.87%	55.16%	+0.71 pp
18	2.624	2.612	2.646	-0.034	61.50%	61.69%	61.11%	+0.59 pp
19	2.402	2.382	2.442	-0.059	63.96%	64.21%	63.48%	+0.73 pp
20	2.962	2.950	2.986	-0.037	54.69%	54.85%	54.35%	+0.50 pp
21	3.292	3.279	3.319	-0.040	51.67%	51.87%	51.29%	+0.58 pp
22	3.594	3.583	3.616	-0.033	44.99%	45.14%	44.68%	+0.47 pp
23	2.850	2.843	2.864	-0.021	43.87%	44.06%	43.50%	+0.56 pp
24	2.838	2.820	2.875	-0.055	42.48%	42.87%	41.70%	+1.17 pp
25	1.876	1.870	1.887	-0.017	66.40%	66.26%	66.69%	-0.42 pp
26	1.657	1.649	1.672	-0.023	69.91%	69.91%	69.90%	+0.01 pp
27	1.938	1.952	1.912	+0.039	73.13%	73.05%	73.30%	-0.25 pp

B. Control Experiment with Silent Audio

This experiment constitutes a control condition for the analysis of attention patterns. The configuration is identical to the main experiment: Qwen2.5-Omni-7B, HumMusQA, 960 evaluations obtained from 320 questions with three permutations of the answer options, the same generation procedure, and the same attention capture. The only difference is that the real audio is replaced, in each evaluation, by a silent WAV file, composed only of zeros at 16 kHz. The objective is to verify whether the attention profile towards audio depends on the acoustic content or on a structural property of the model.

With silent audio, the model reaches an accuracy of 48.65%, corresponding to 467 correct answers out of 960. This value is above the chance level, equal to 25%, but below the accuracy obtained with real audio, equal to 66.46%. The drop of 17.81 percentage points suggests that real audio provides useful information; at the same time, the above-chance performance confirms that many answers remain partially inferable from the textual component alone.

Table 10: Accuracy with real audio, silent audio, and chance level.

Condition	Accuracy	N
Real audio	66.46%	960
Silent audio	48.65%	960
Chance, 4 options	25.00%	—

Semantic consistency across permutations is equal to 67.39%, with 31 questions out of 46 in which the model

selects the same semantic option across all three permutations. Even in the absence of informative audio, therefore, the model’s behaviour remains relatively stable.

The layer-by-layer attention profile remains similar to that observed with real audio. Audio receives the largest share in the early stages of the network, but becomes almost negligible in the intermediate and deep layers. Text stably dominates throughout the full depth of the model.

Table 11: Mean attention towards audio and text across the three network regions with silent audio.

Region	Layer	Audio	Overall text
Early	L0–L9	3.84%	96.16%
Middle	L10–L18	1.23%	98.77%
Late	L19–L27	1.01%	98.99%

Some representative layers confirm the same dynamic: attention to text increases until it exceeds 99% in the final layers, while silent audio falls below 1%.

Table 12: Representative examples of attention by layer with silent audio.

Layer	Audio	Instruction	Query text
Layer 0	5.50%	52.20%	94.50%
Layer 13	1.10%	30.30%	98.90%
Layer 27	0.72%	16.20%	99.30%

The category-level analysis shows strong variability. Sound Texture, Genre and Style, and Functional Context maintain high accuracy even with silent audio, suggesting greater textual solvability. By contrast, Musical Texture, Lyrics, and Melody are more penalised, indicating greater dependence on real acoustic information.

Table 13: Accuracy by musical category with silent audio.

Category	Accuracy	N
Sound Texture	70.37%	81
Genre and Style	59.09%	66
Functional Context	56.25%	48
Historical/Cultural	51.28%	39
Mood and Expression	50.79%	63
Harmony	50.00%	108
Metre and Rhythm	44.12%	102
Instrumentation	43.39%	189
Structure	42.86%	21
Melody	42.16%	102
Lyrics	39.39%	33
Musical Texture	36.36%	33

Difficulty also produces the expected trend: performance progressively decreases from Low to Medium and High questions.

Table 14: Accuracy by difficulty level with silent audio.

Difficulty	Correct	Total	Accuracy
Low	221	426	51.88%
Medium	174	369	47.15%
High	72	165	43.64%

The experiment leads to two main indications. First, real audio is informative: replacing it with silence reduces accuracy by 17.81 percentage points. Second, the attention pattern remains almost unchanged: even with content-free audio, the model assigns only a minimal share of attention to audio and maintains a strong textual dominance.

These results clarify the relationship with attention knockout. The fact that direct audio knockout produces very small effects does not imply that audio is useless; rather, it indicates that its information does not emerge primarily as direct attention from the decision token towards the audio tokens. It is more plausible that the acoustic signal is integrated in a distributed manner during the prefill and in the construction of the multimodal context.

In summary, real audio contributes to the response, but the observable attentional profile remains structurally text-first. The experiment therefore confirms that the presence of audio, attention towards audio, and the causal role of audio are distinct dimensions that must be analysed separately.

Abstract in lingua italiana

I Large Audio Language Models stanno diventando sempre più efficaci nel rispondere a domande sulla musica, ma una risposta corretta non dimostra, di per sé, che stiano realmente ascoltando. Un modello può raggiungere la risposta giusta fondando la propria decisione sul segnale acustico, oppure affidandosi a prior testuali, conoscenze teorico-musicali e bias appresi durante il pretraining. Questa tesi indaga se Qwen2.5-Omni-7B utilizzi effettivamente il suono nella risoluzione di task di Music Question Answering, o se le sue decisioni siano guidate principalmente dal testo.

Per rispondere a questa domanda, il lavoro sviluppa una pipeline di interpretabilità per il music understanding multimodale. MM-SHAP viene adattato per misurare il contributo relativo di audio e testo, mentre audioDIME viene introdotto come variante di DIME per il dominio audio, capace di separare effetti unimodali e interazioni multimodali. Il metodo combina scoring basato sui logits, source separation tramite Demucs, segmentazione temporale onset-based e perturbazioni ispirate ad audioLIME, rappresentando l'audio attraverso coppie sorgente-segmento interpretabili.

Gli esperimenti su HumMusQA rivelano un chiaro bias text-first. Il testo risulta altamente sufficiente e necessario per preservare le decisioni del modello, mentre i contributi puramente audio sono spesso sostituibili. L'audio diventa rilevante soprattutto all'interno delle interazioni audio-testo, suggerendo che il suo ruolo causale sia mediato dalla domanda più che agire come segnale percettivo indipendente.

Le analisi interne del *Thinker*, basate su attention patterns, Logit Lens e attention knockout, rafforzano questa interpretazione. Il modello processa strutturalmente i token audio, ma la loro influenza causale diretta sulla risposta finale è debole. Bloccare il canale audio diretto produce effetti marginali, mentre interrompere la più ampia struttura attentiva testo-audio nei layer intermedi risulta più dannoso. Nel complesso, questa tesi sostiene che la valutazione degli Audio LLMs richieda di distinguere tra multimodalità strutturale e ascolto realmente fondato sul segnale acustico, fornendo strumenti per analizzare non solo se un modello risponde correttamente, ma come tale risposta viene prodotta.

Parole chiave: Comprensione Musicale, Audio Language Models, Intelligenza Artificiale Spiegabile, Interpretabilità Multimodale.

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Che questa tesi sia di monito per i futuri studenti laureandi. Come tanti, anche io ho pensato che questo percorso fosse per me troppo complesso, che io non fossi adatto ad esso. Ma pensare a quel ragazzo che a Settembre 2019 ha varcato i cancelli dell'Università di Roma La Sapienza, è entrato in un'aula con quattrocento persone, si è seduto ed ha ascoltato con timore la prima lezione, che col tempo ha sognato e bramato sempre più questo ultimo momento: eccoti, Flavio, sei un uomo indipendente, hai avuto il coraggio di cambiare vita molto giovane, hai riempito la valigia di esperienze e persone, ti sei dimenato tra le difficoltà che l'Università ti ha posto davanti, a volte svogliatamente, a volte con merito, ma mai con fortuna o scorciatoie. Sii fiero di te e del tuo impegno, anche rileggendo queste righe tra molti anni, perchè questo sei Tu, alla fine ce l'hai fatta e brami ad un obiettivo ancora più ambizioso ora. Non smettere mai di guardare il mondo con gli occhi di un bambino, sperimenta, crea, gioca, divertiti, pensa, anche se ti taceranno di essere infantile. La tua vita te la comandi tu, anche se a volte ti senti come un veliero in bonaccia.